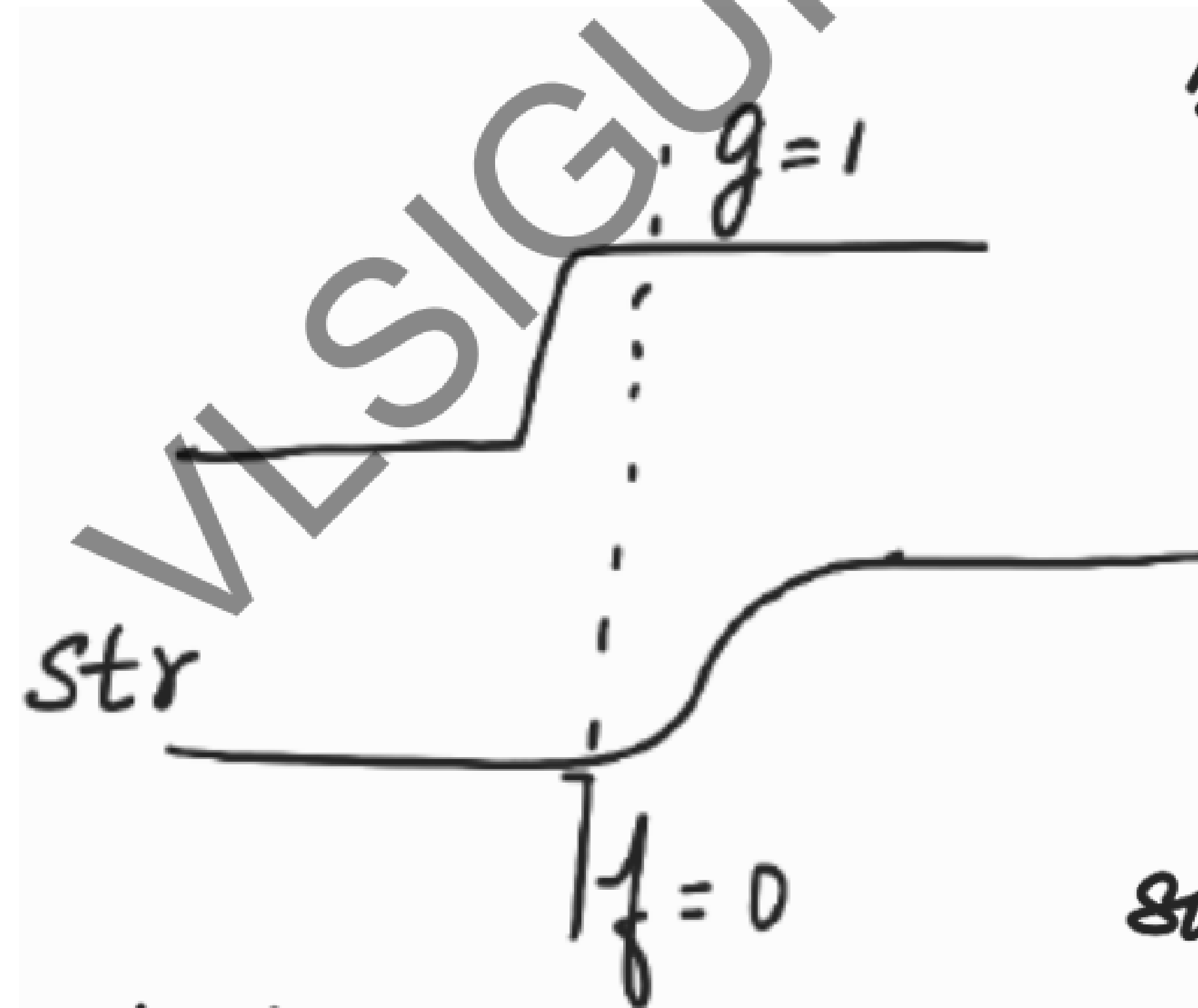


TRANSITION DELAY FAULT MODEL (TDF)

Part 1

TDF fault model is used to detect that the logic level from 0-1 and 1-0 properly transit within the specific time (time period of fast frequency clock) or not.



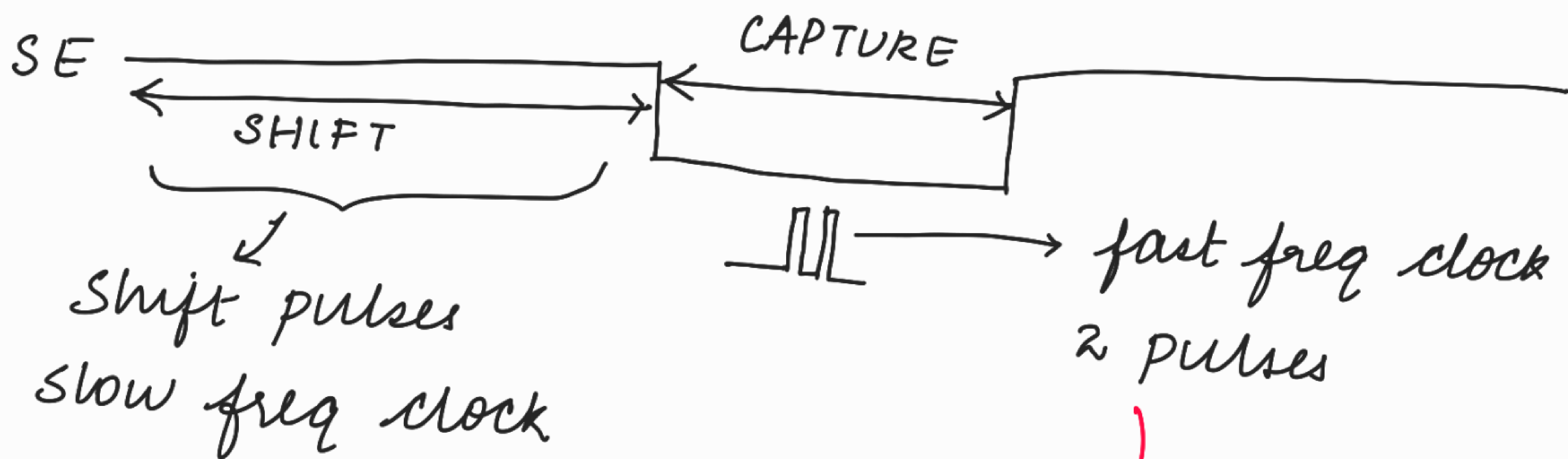
Why fast freq clk?

Ans: Because after manufacturing (i.e., after testing when the chip goes to the customer) only fast freq clk will be used.

So we need to check

whether the rise/fall of signals happens within the time period of fast freq clk or not.

TRANSITION DELAY FAULT MODEL :-

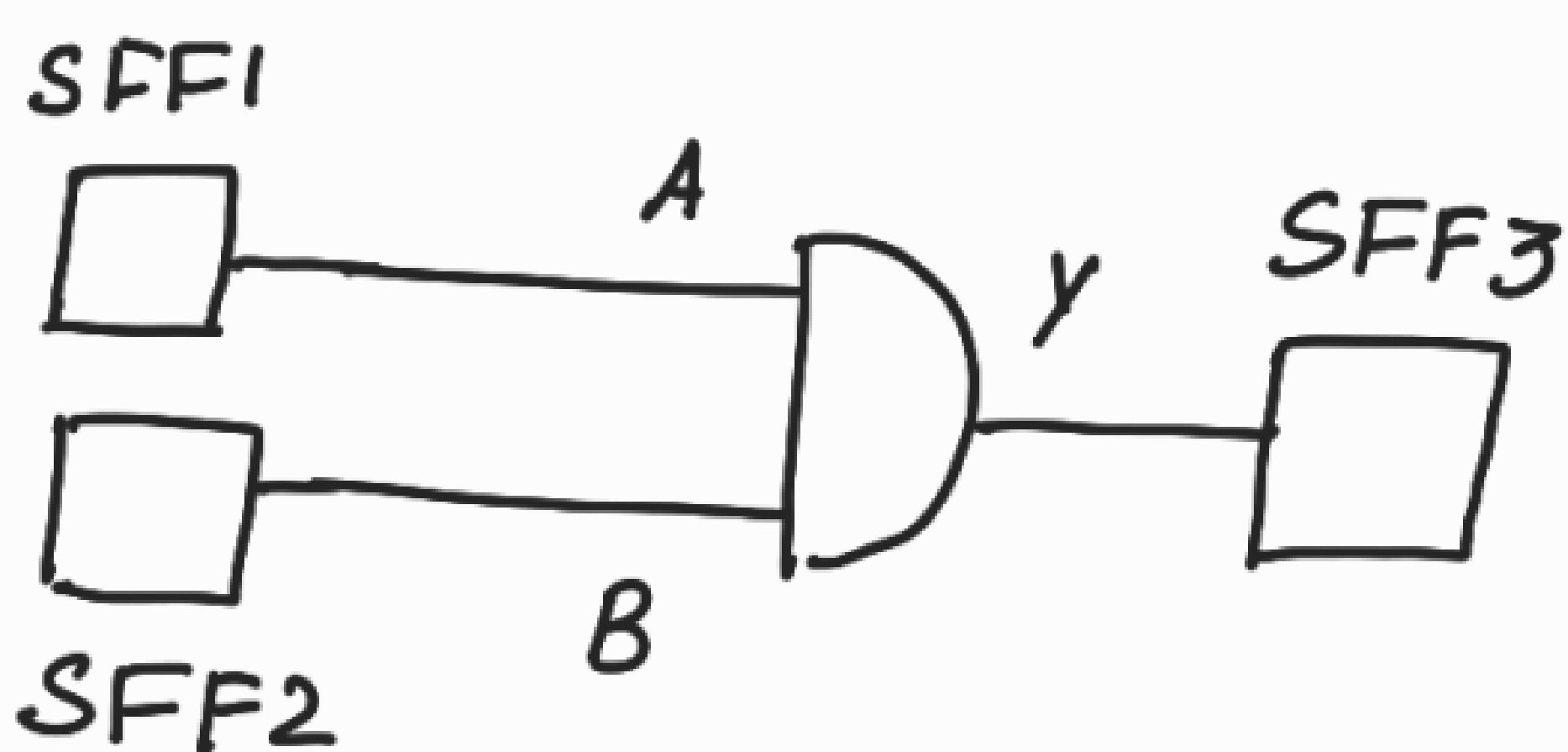


because TDF is done to check whether the rise/fall of signals happens within the time period of fast freq clock or not.

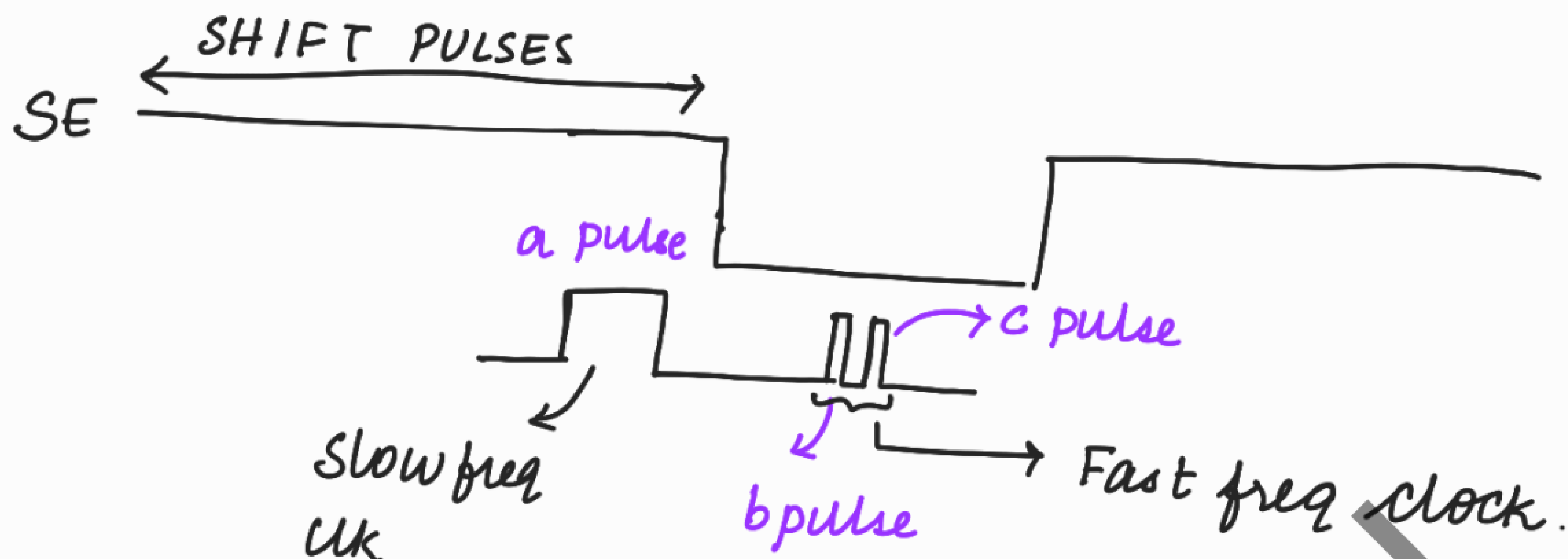


We need 2 clock pulses because we need to launch and capture the transition

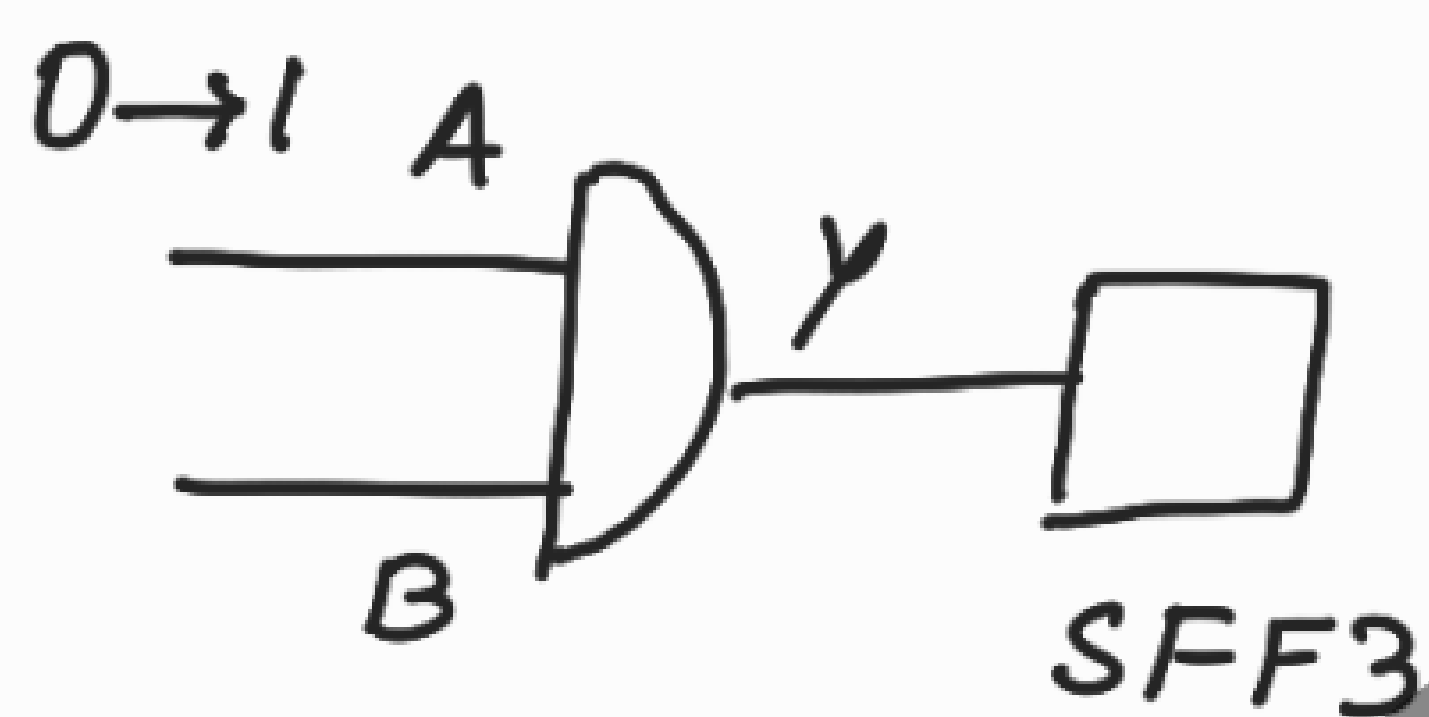
Eg 1:



str : slow to rise
stf : slow to fall



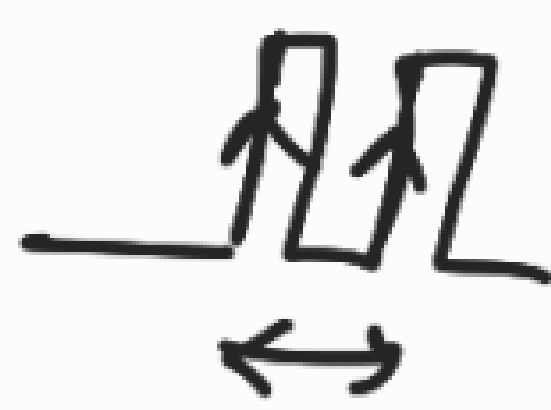
A str fault :-



Tool will try to bring a 0-1 transition on netA and propagate the value to an observation point (SFF3)

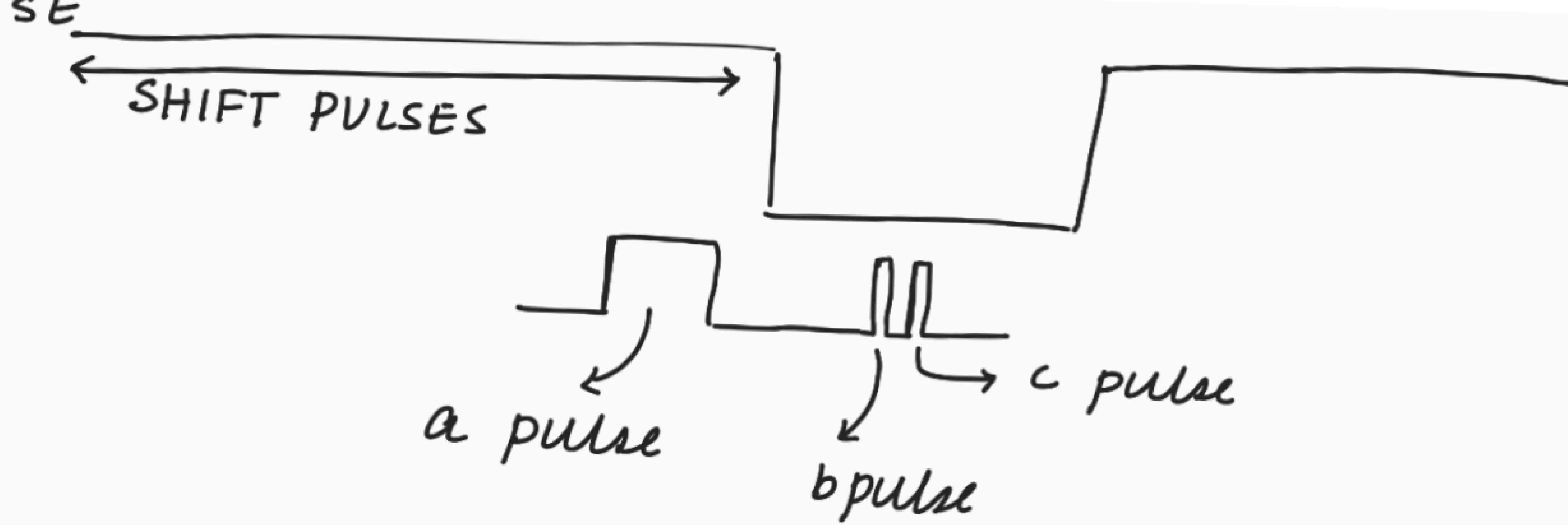
After c pulse, the tool will observe the value in the observation point (SFF3)

If SFF3 holds 0, then there is slow to rise fault

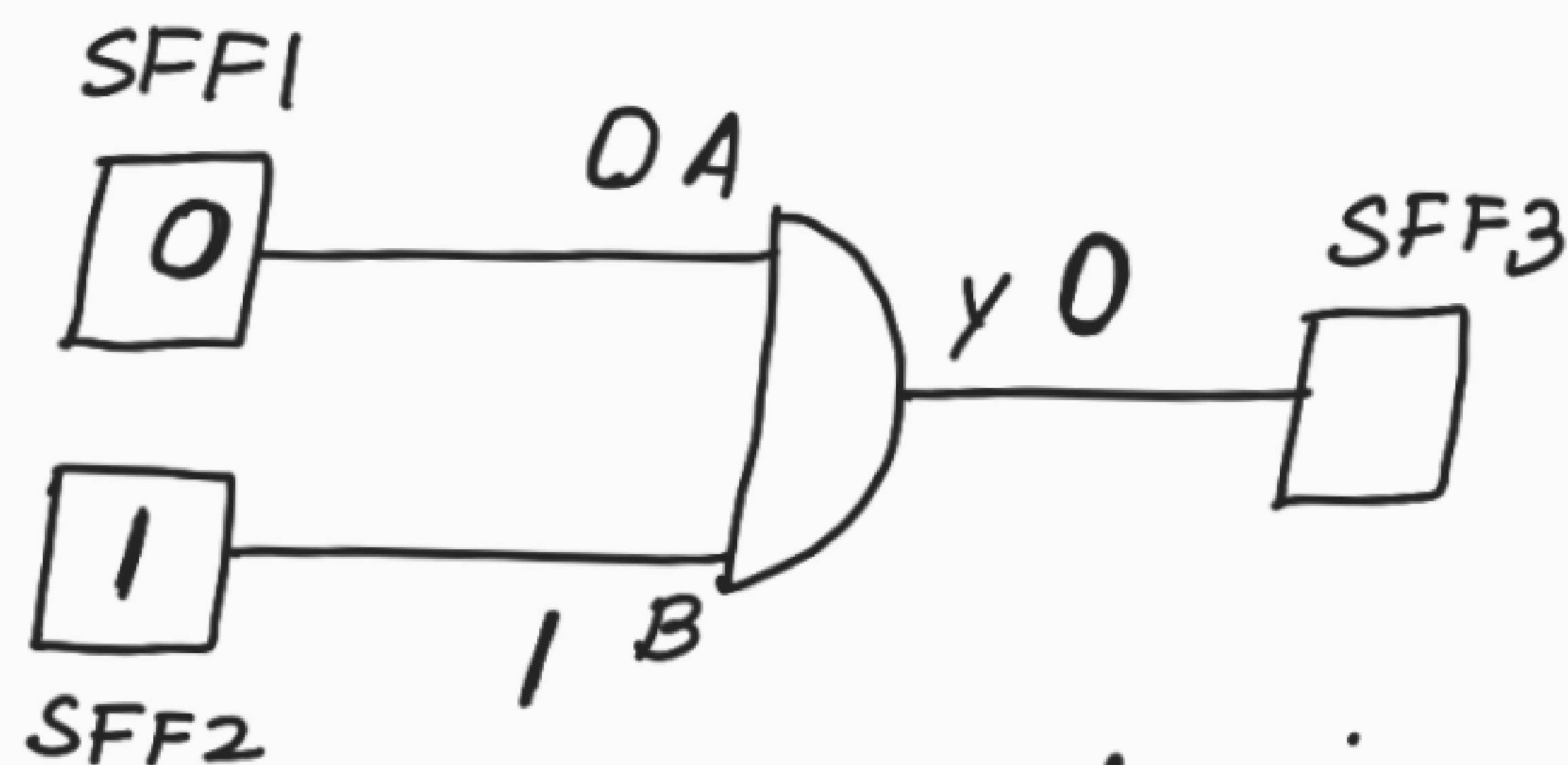


↳ The signal has not risen within this time.

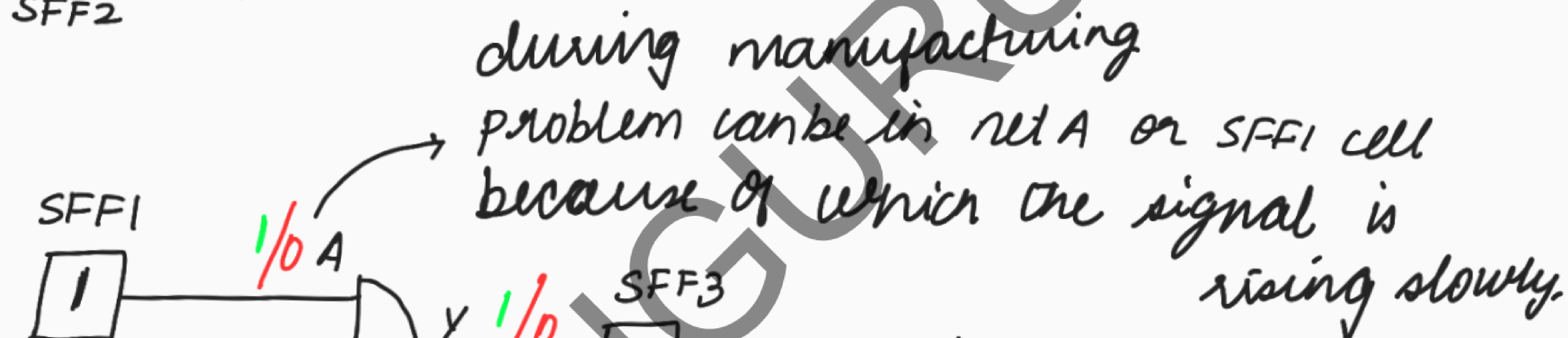
If SFF3 holds 1, then there is no slow to rise fault



A slow to rise fault :- A str

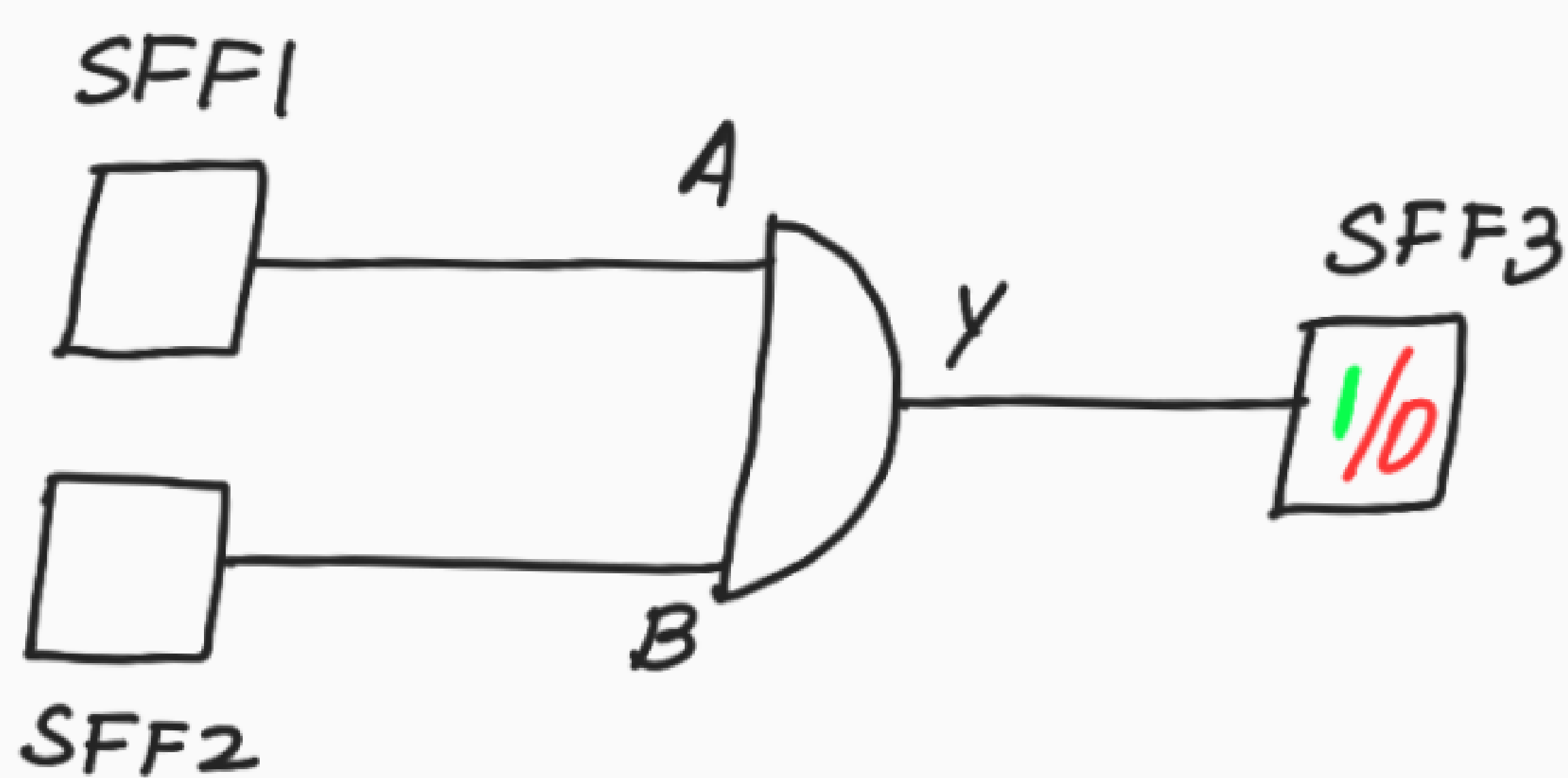


a pulse



b pulse During ATPG.

time, the tool is just modelling or mimicing that faulty behaviour [in faulty m/c]



c pulse same concept as discussed during SA

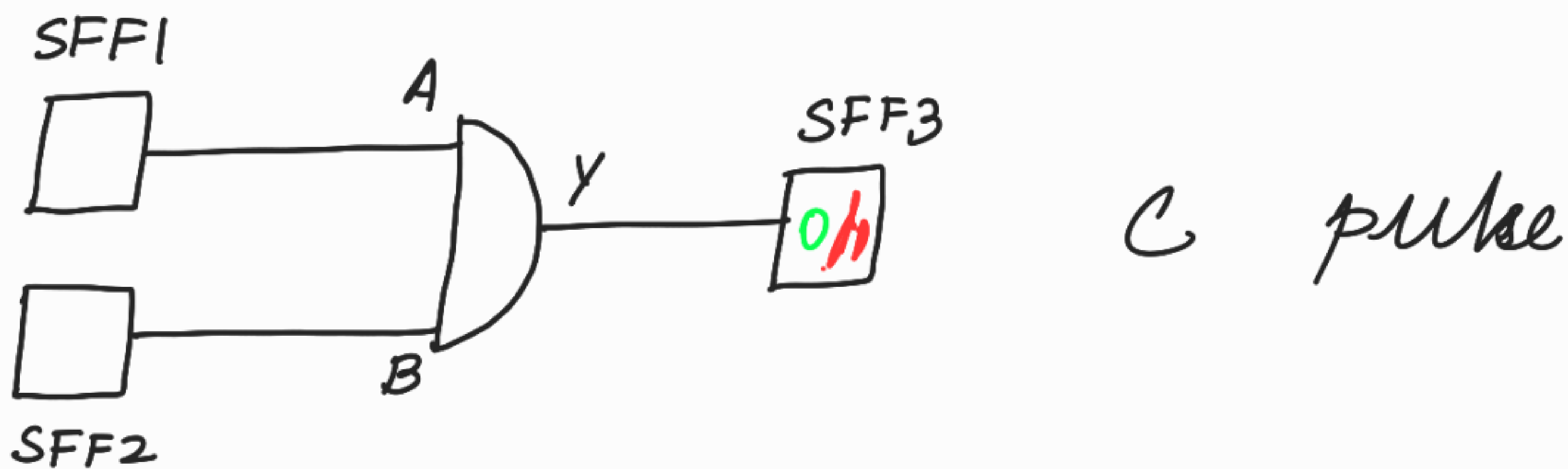
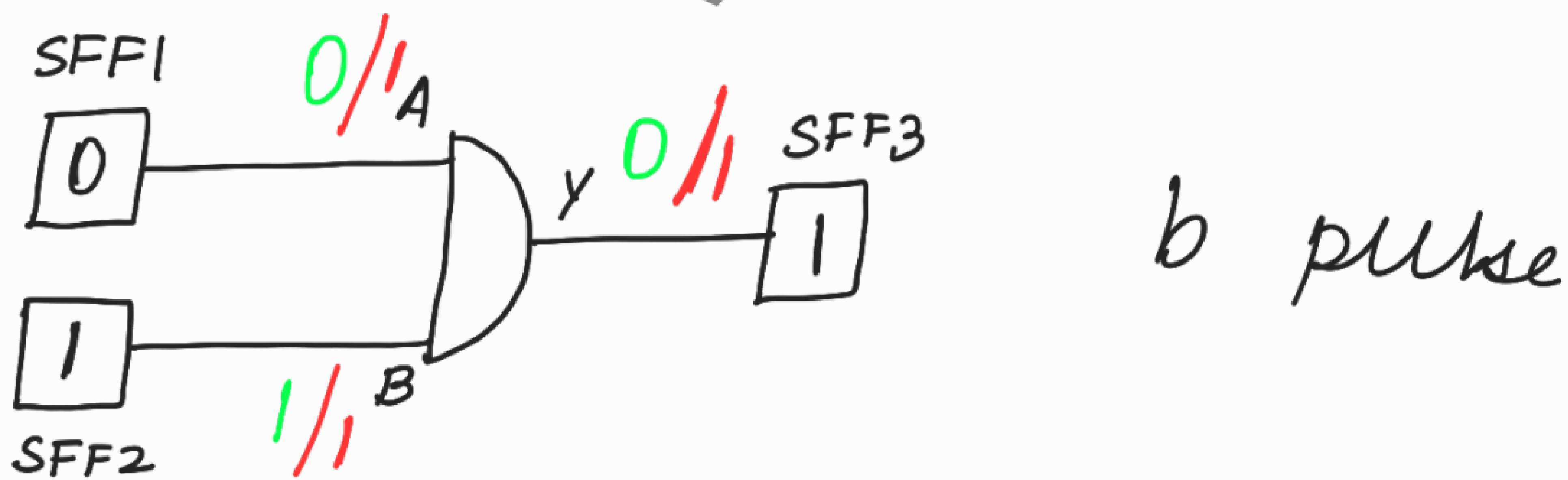
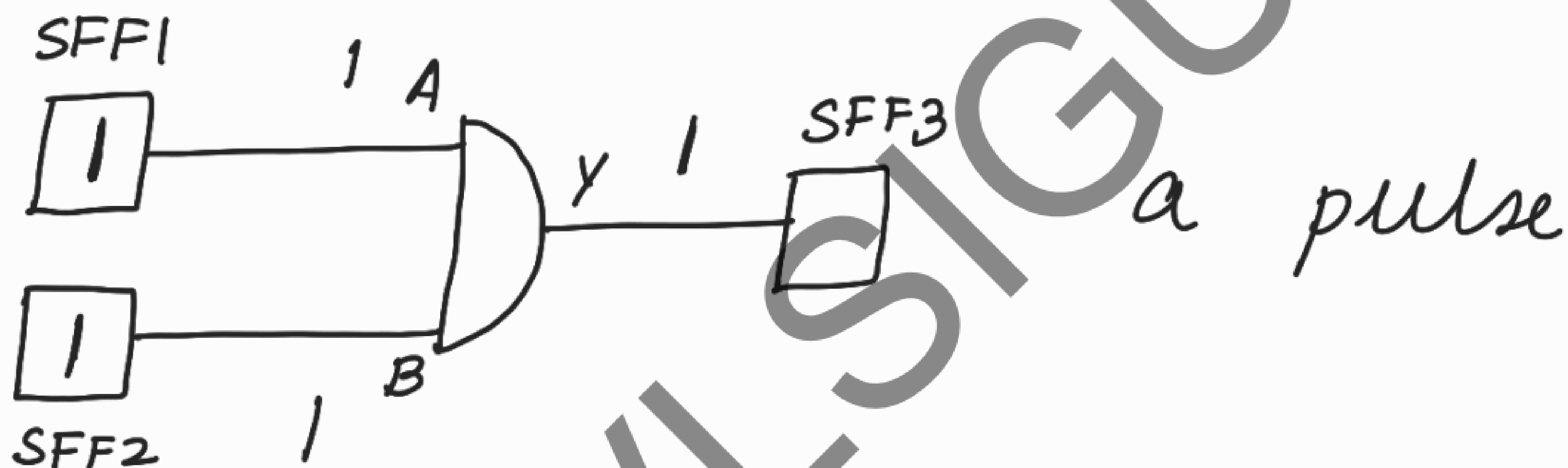
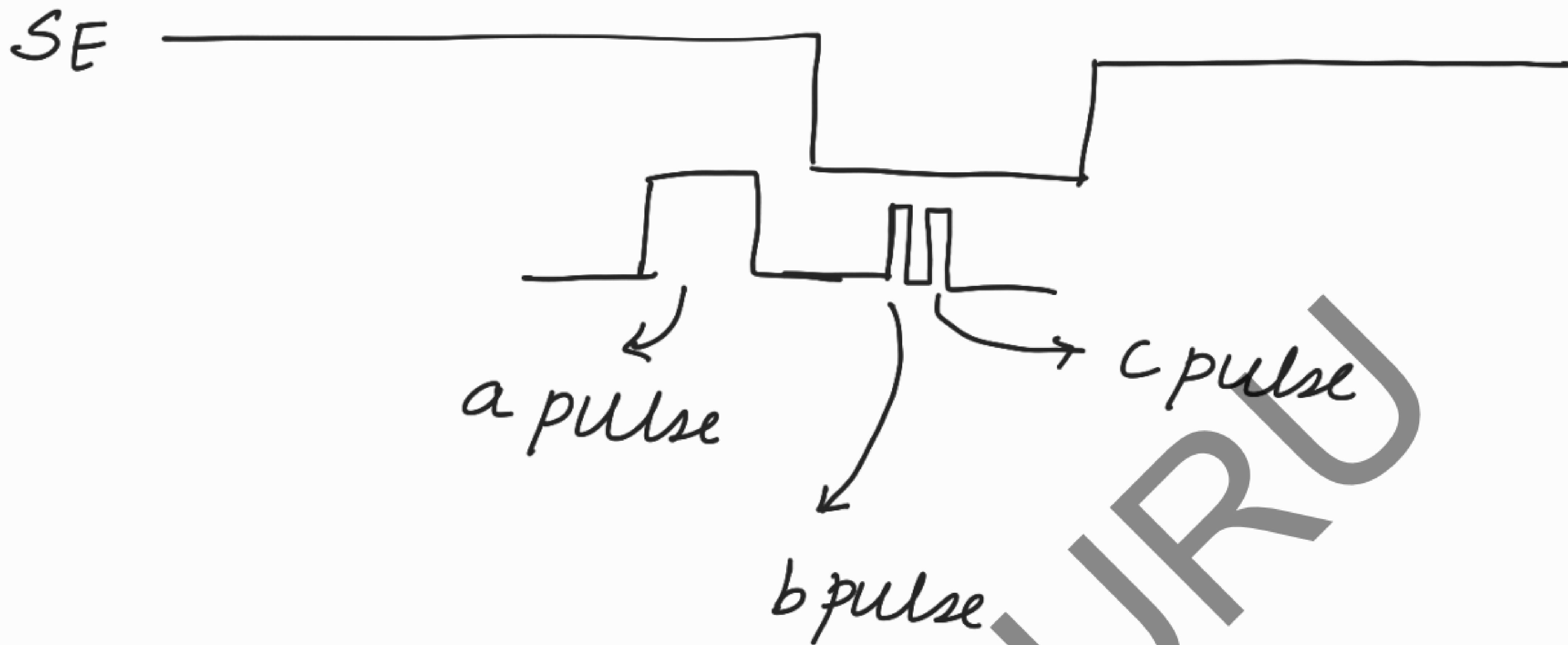
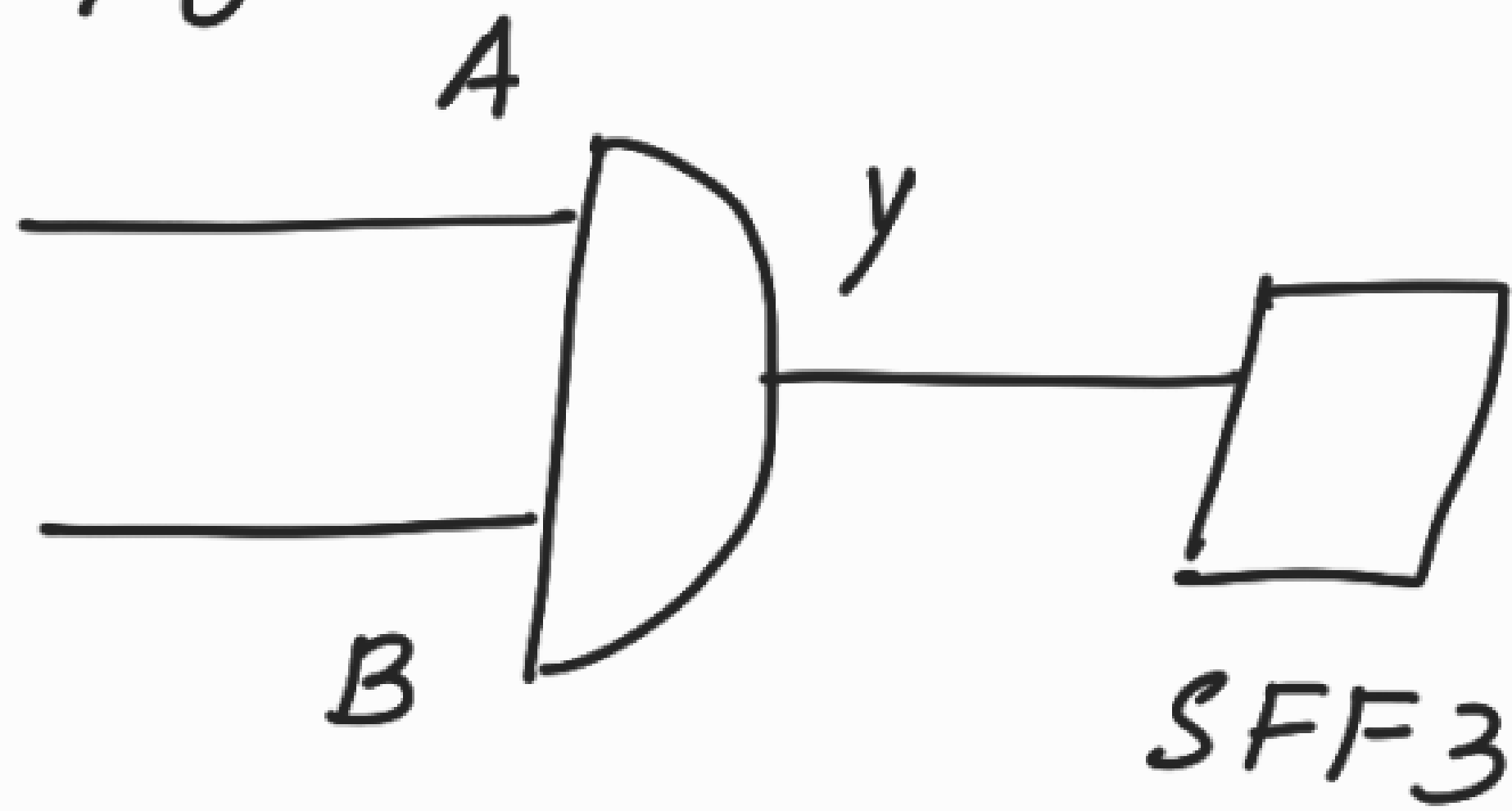
fault model

pattern:-

		A	B
{	A B	0	1
		1	1

A slow to fall (A stf) :

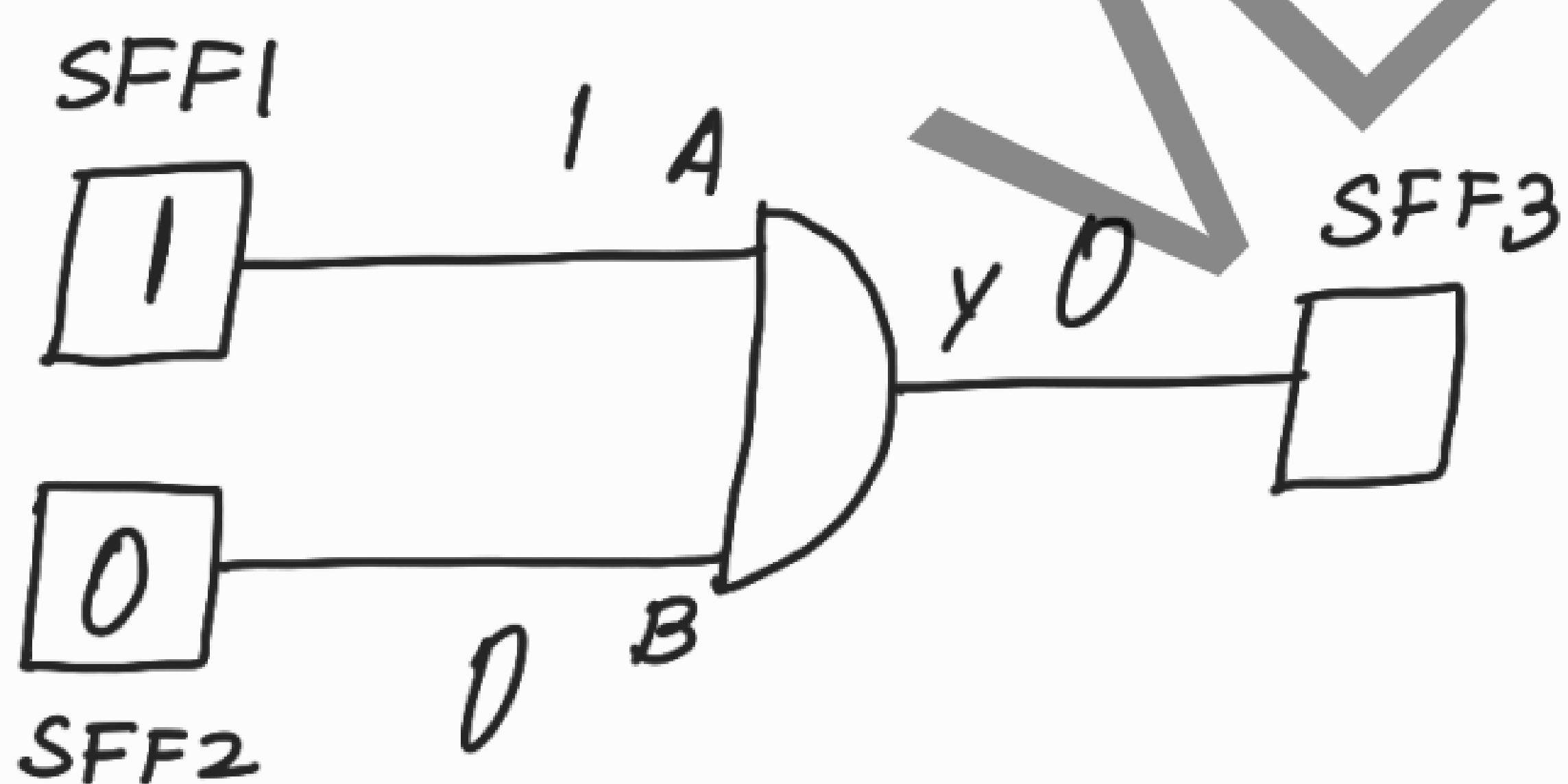
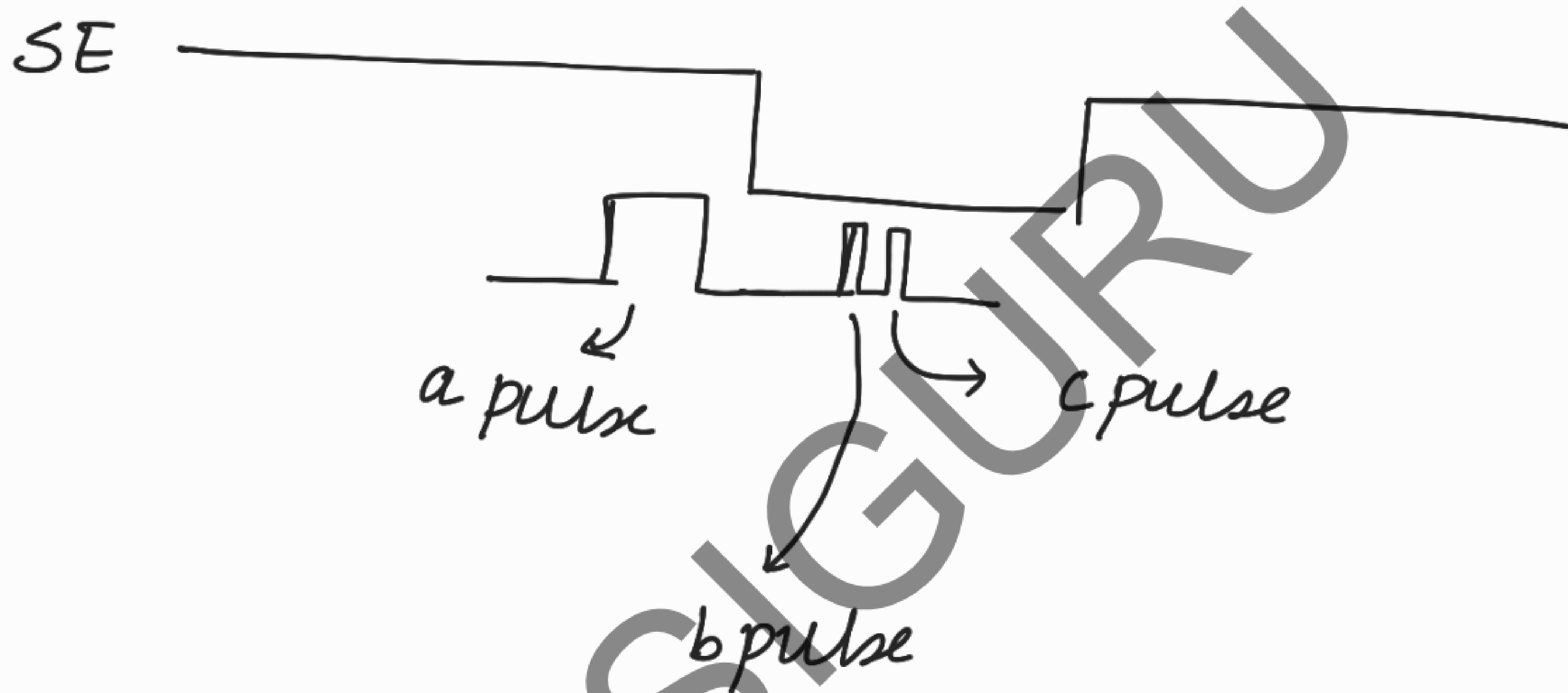
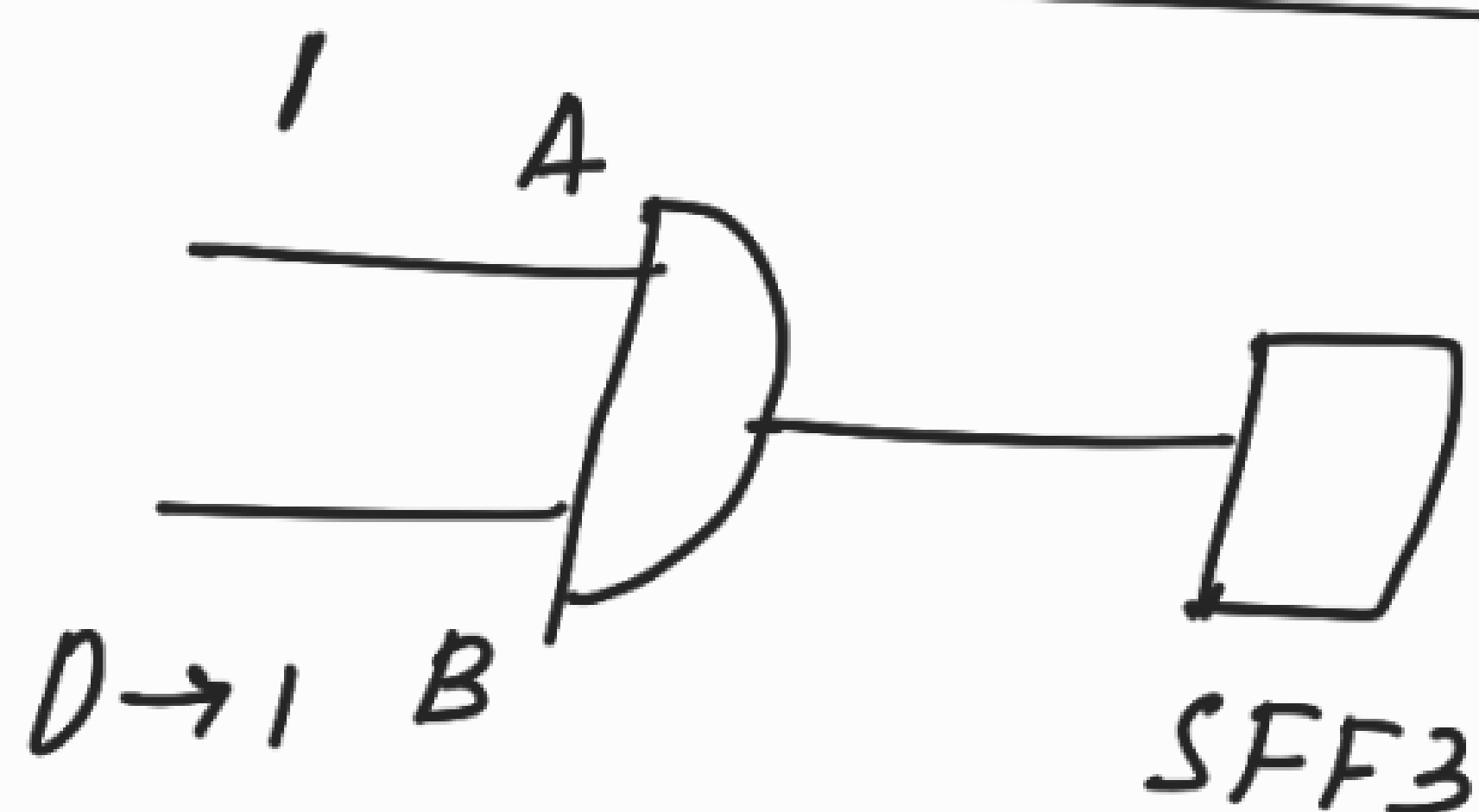
1 → 0



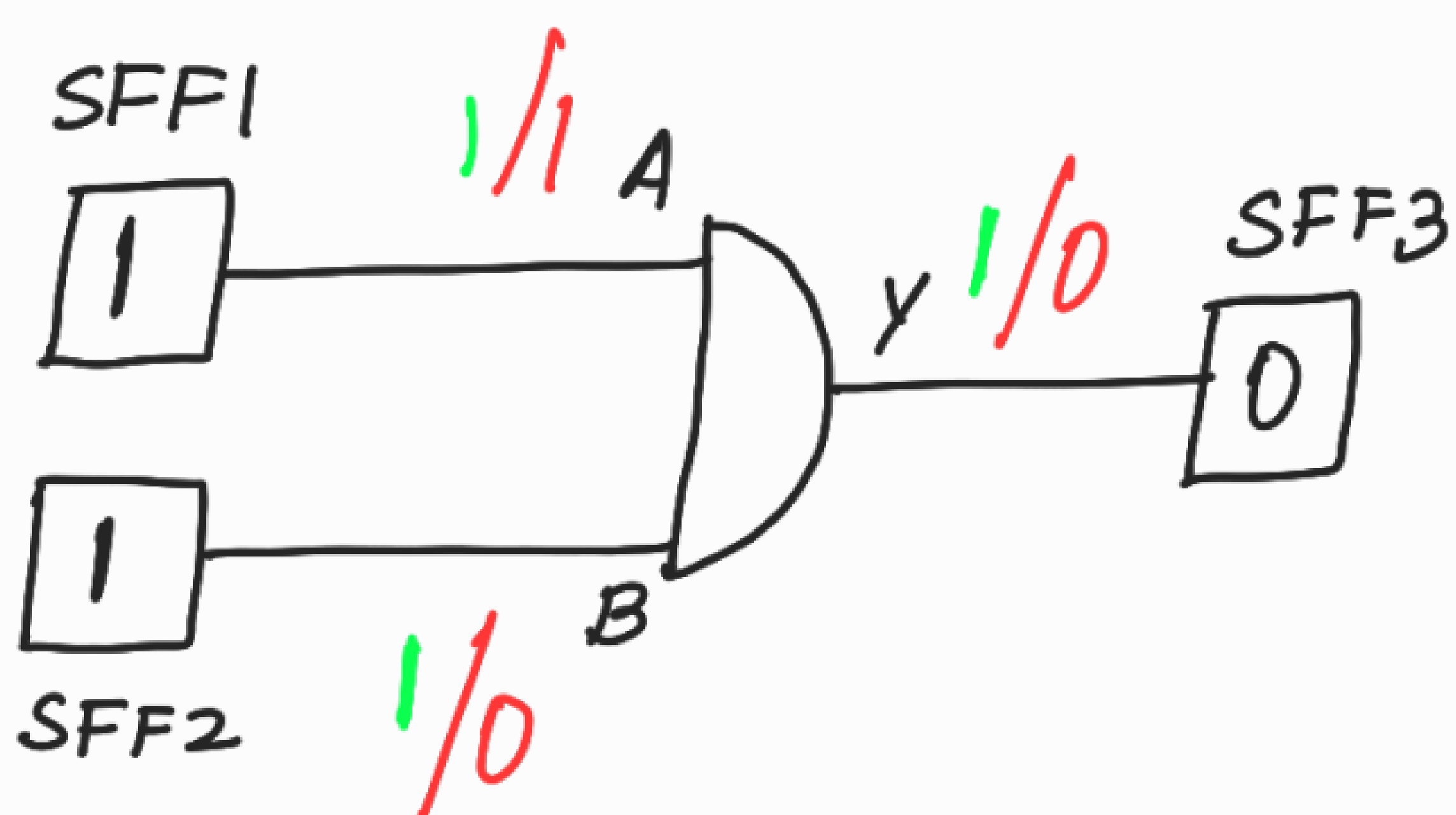
pattern :-

		A	B
AB	{	1	1
		0	1

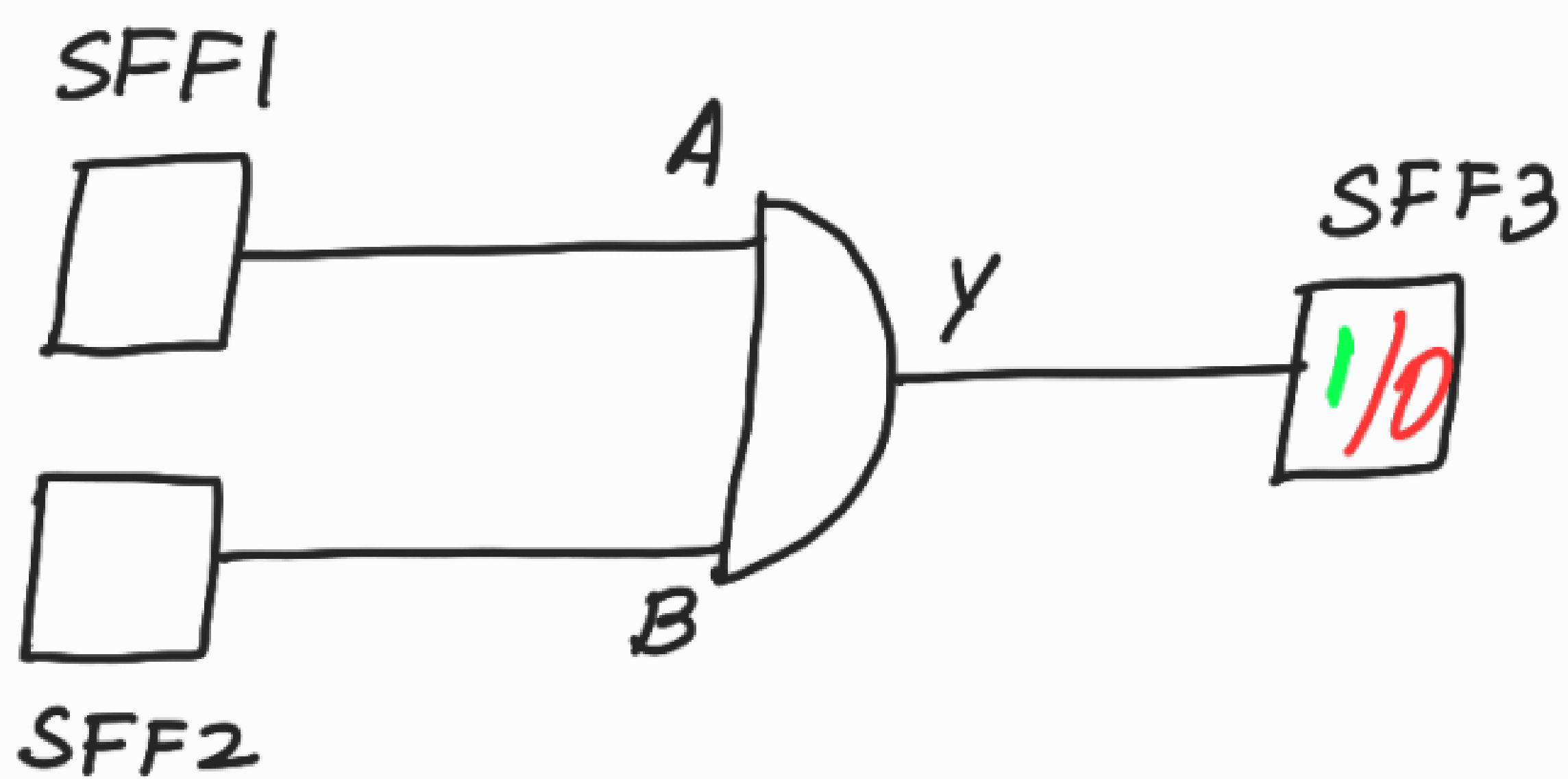
B slow to rise (Bstr) :-



a pulse



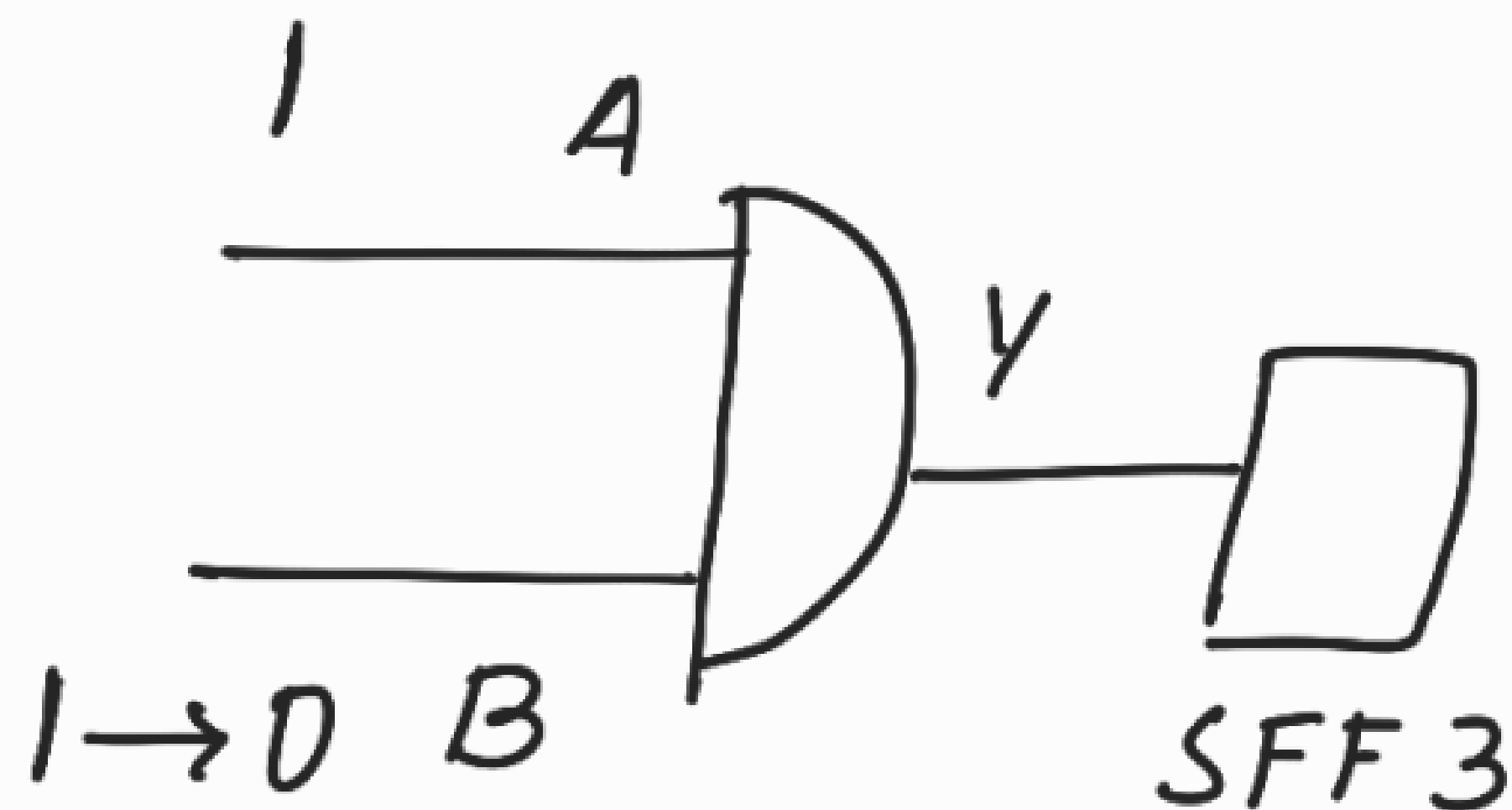
b pulse



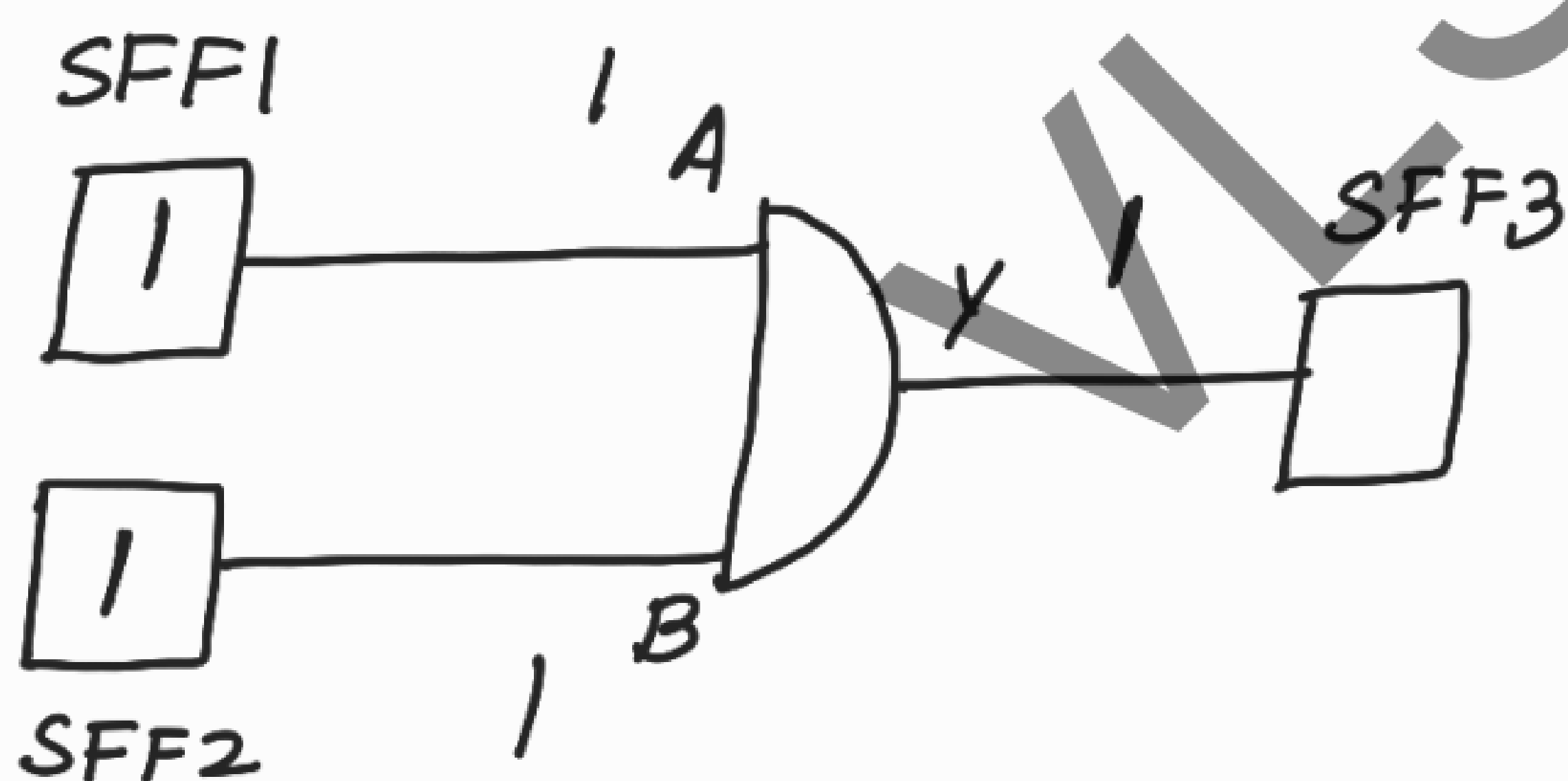
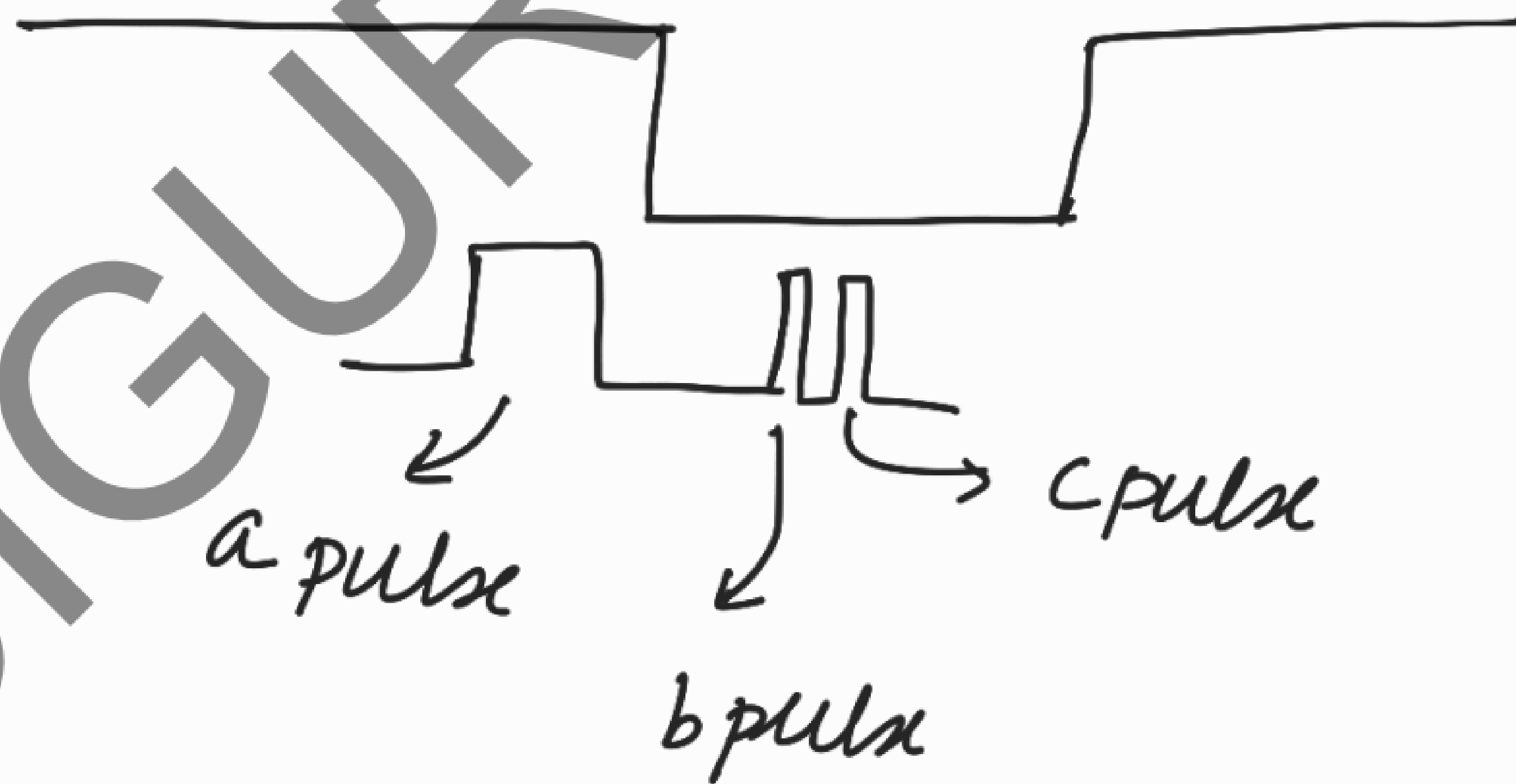
c pulse

pattern:- $AB \left\{ \begin{array}{cc} A & B \\ 1 & 0 \\ 1 & 1 \end{array} \right\}$

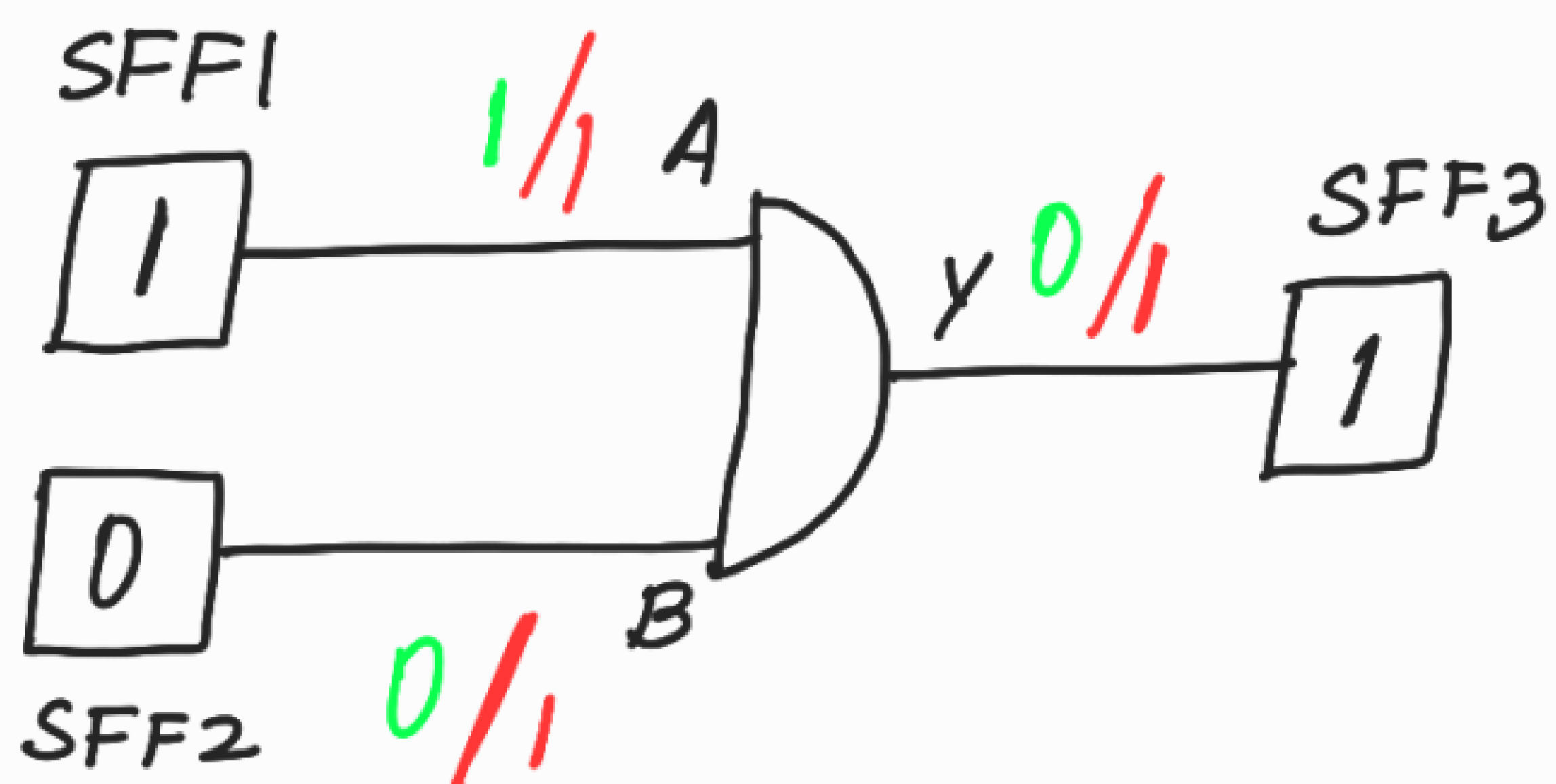
B slow to fall (B stf) :-



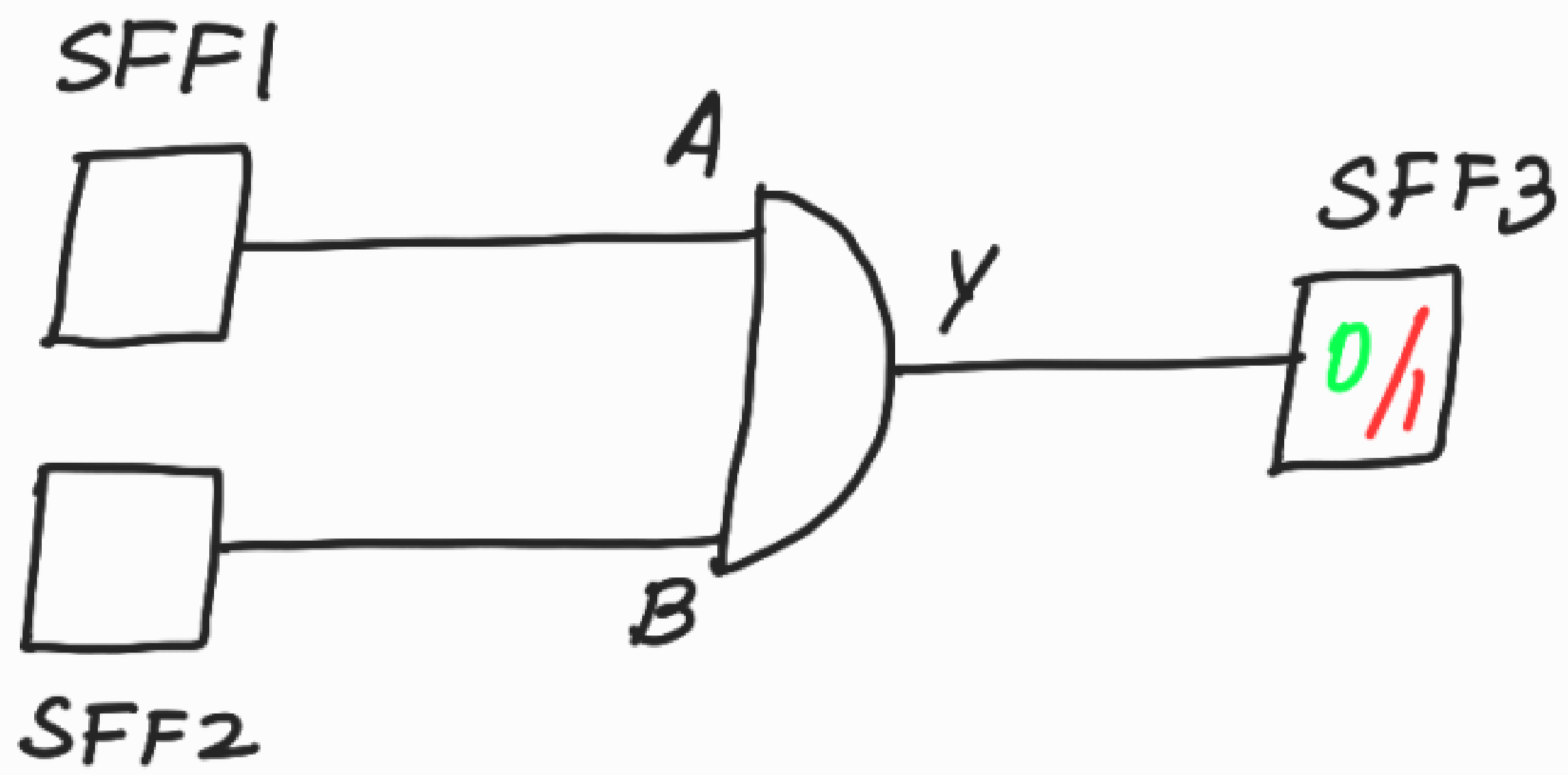
SE



a pulse



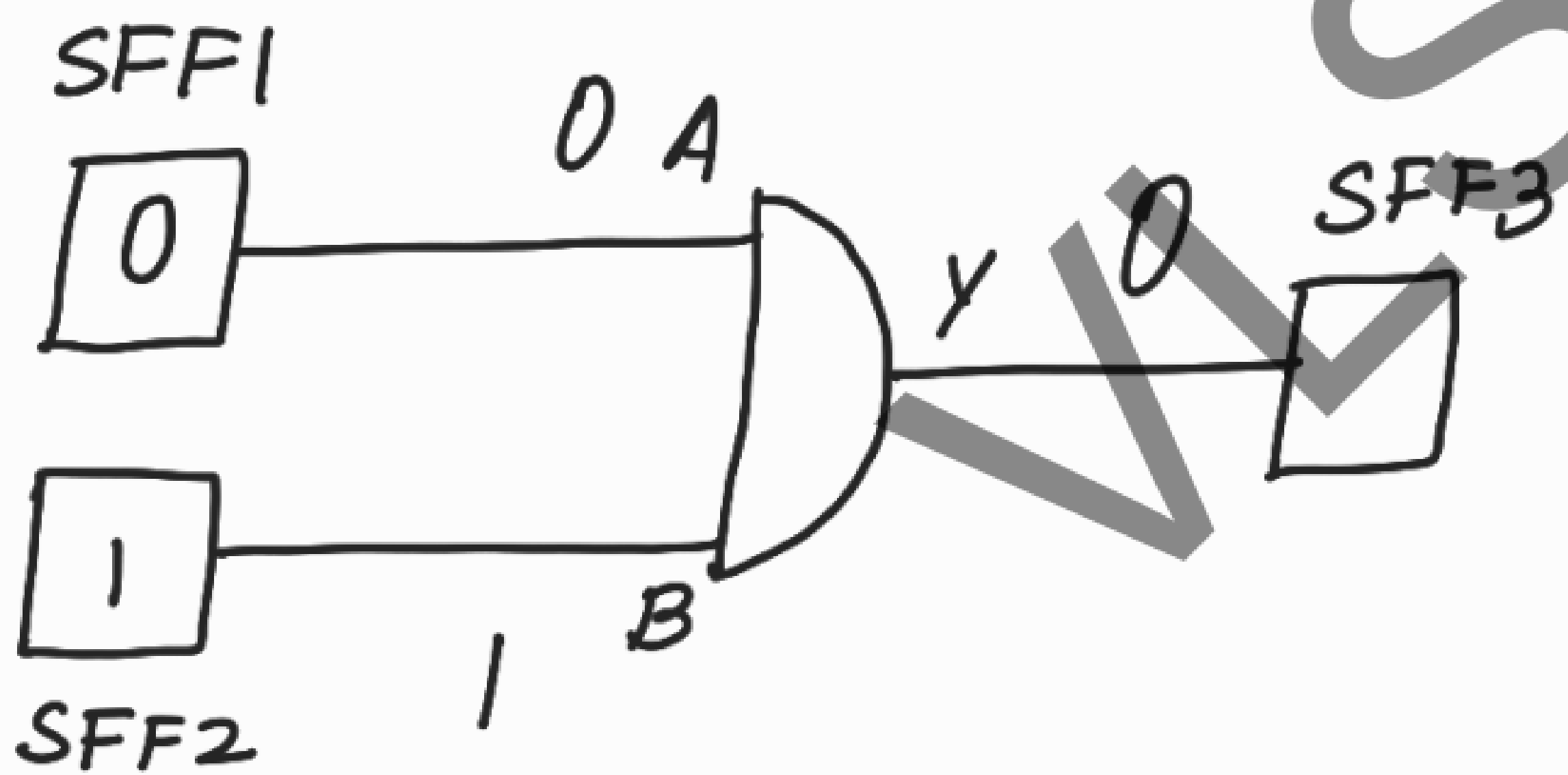
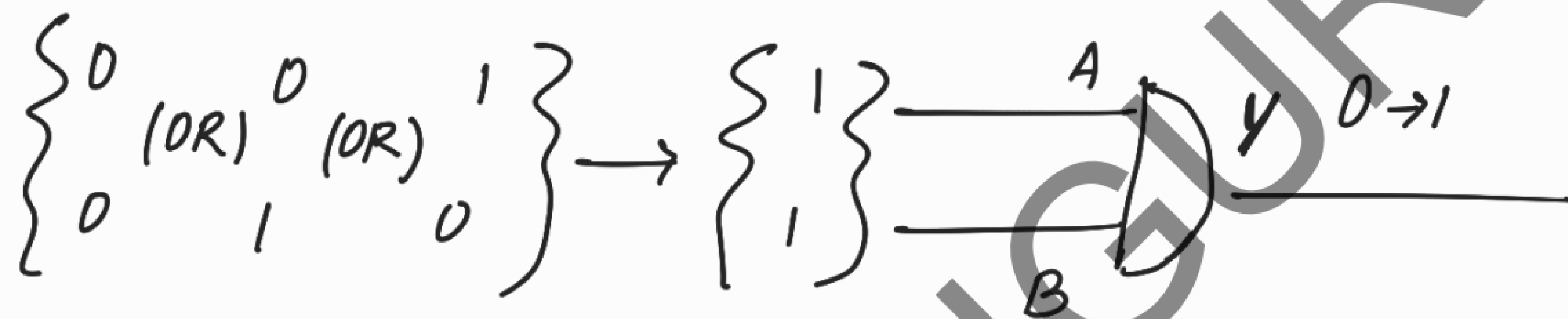
b pulse



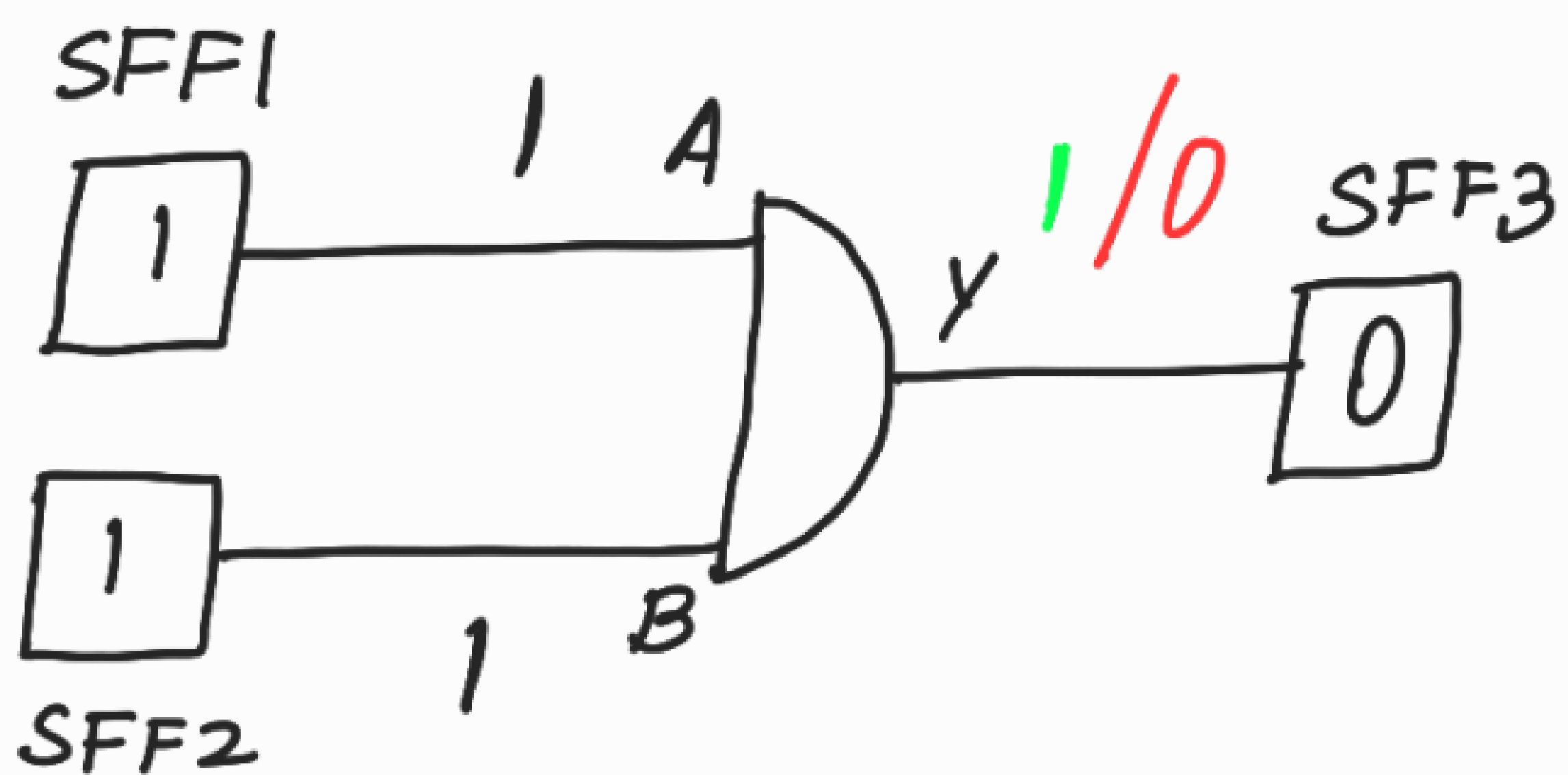
c pulse

pattern:- AB { A B
1 1
1 0 }

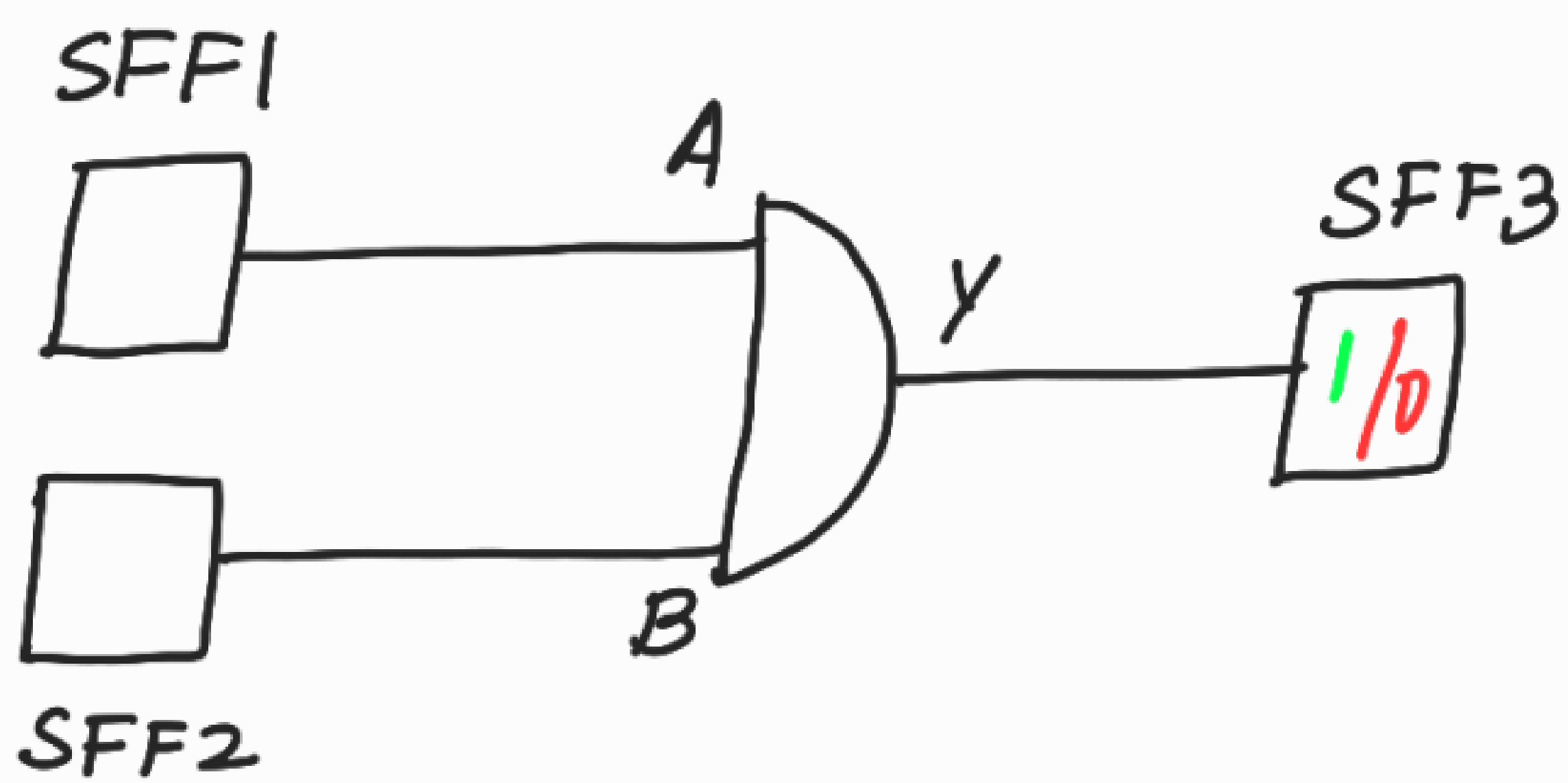
Y slow to rise Y (sta):-



a pulse



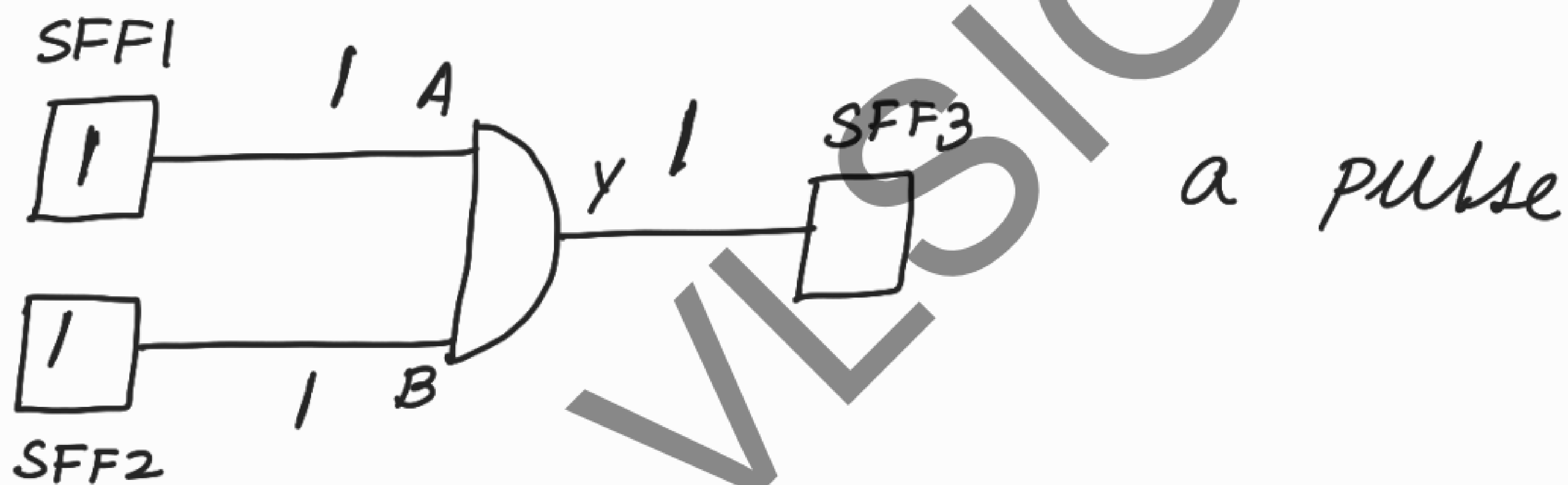
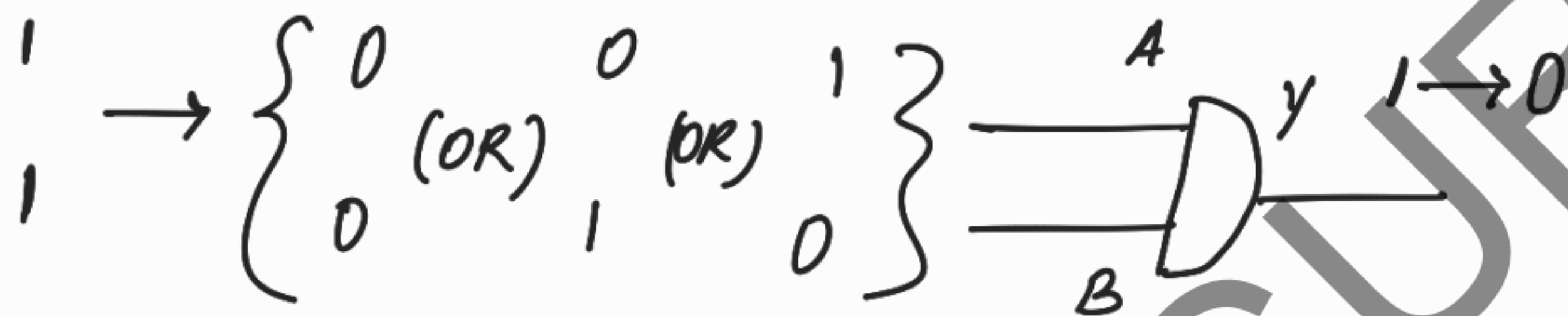
b pulse



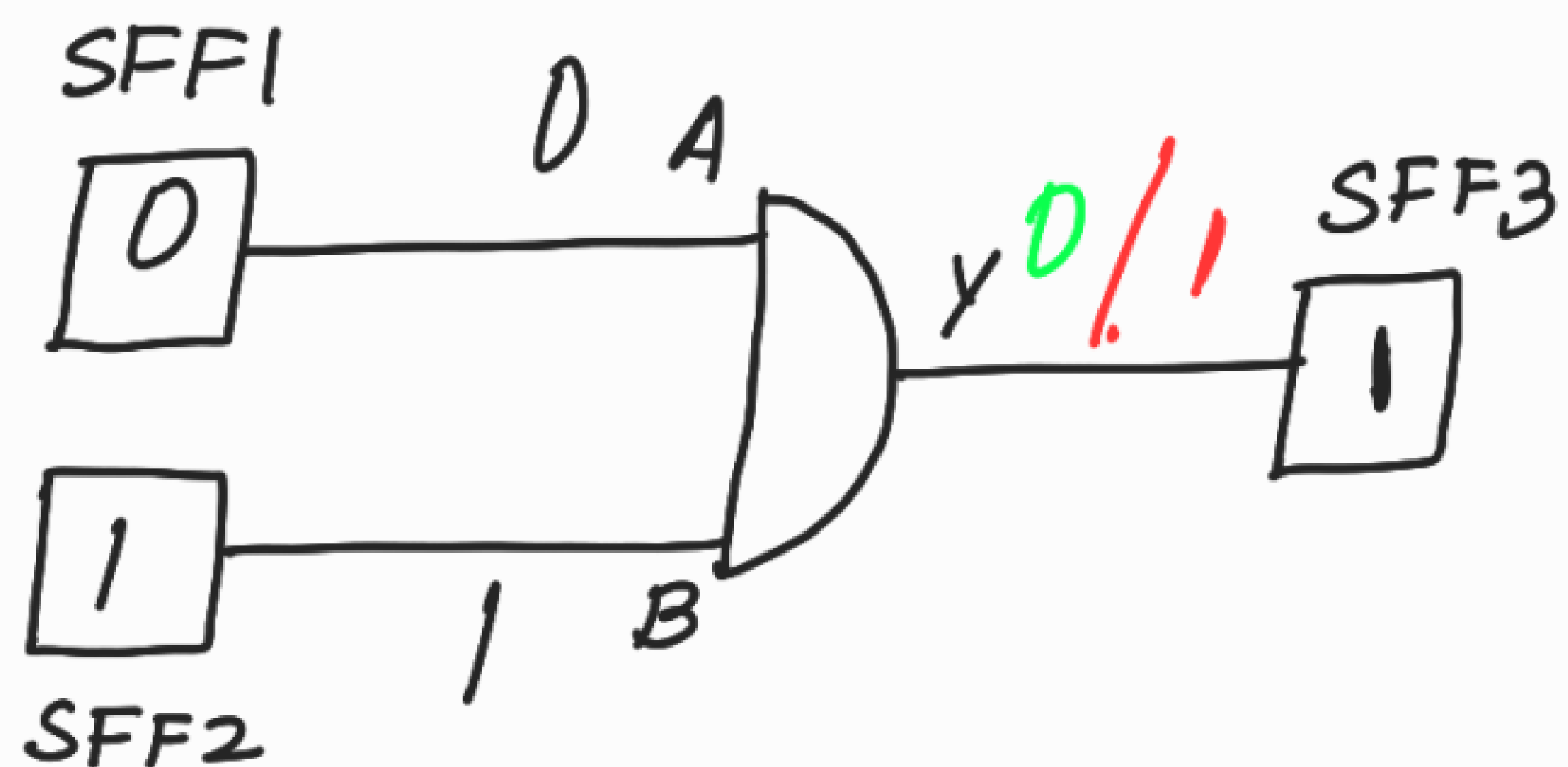
c pulse

pattern:- $AB \left\{ \begin{array}{l} 00 \\ 11 \end{array} \right. (OR) \left\{ \begin{array}{l} 01 \\ 11 \end{array} \right. (OR) \left\{ \begin{array}{l} 10 \\ 11 \end{array} \right.$

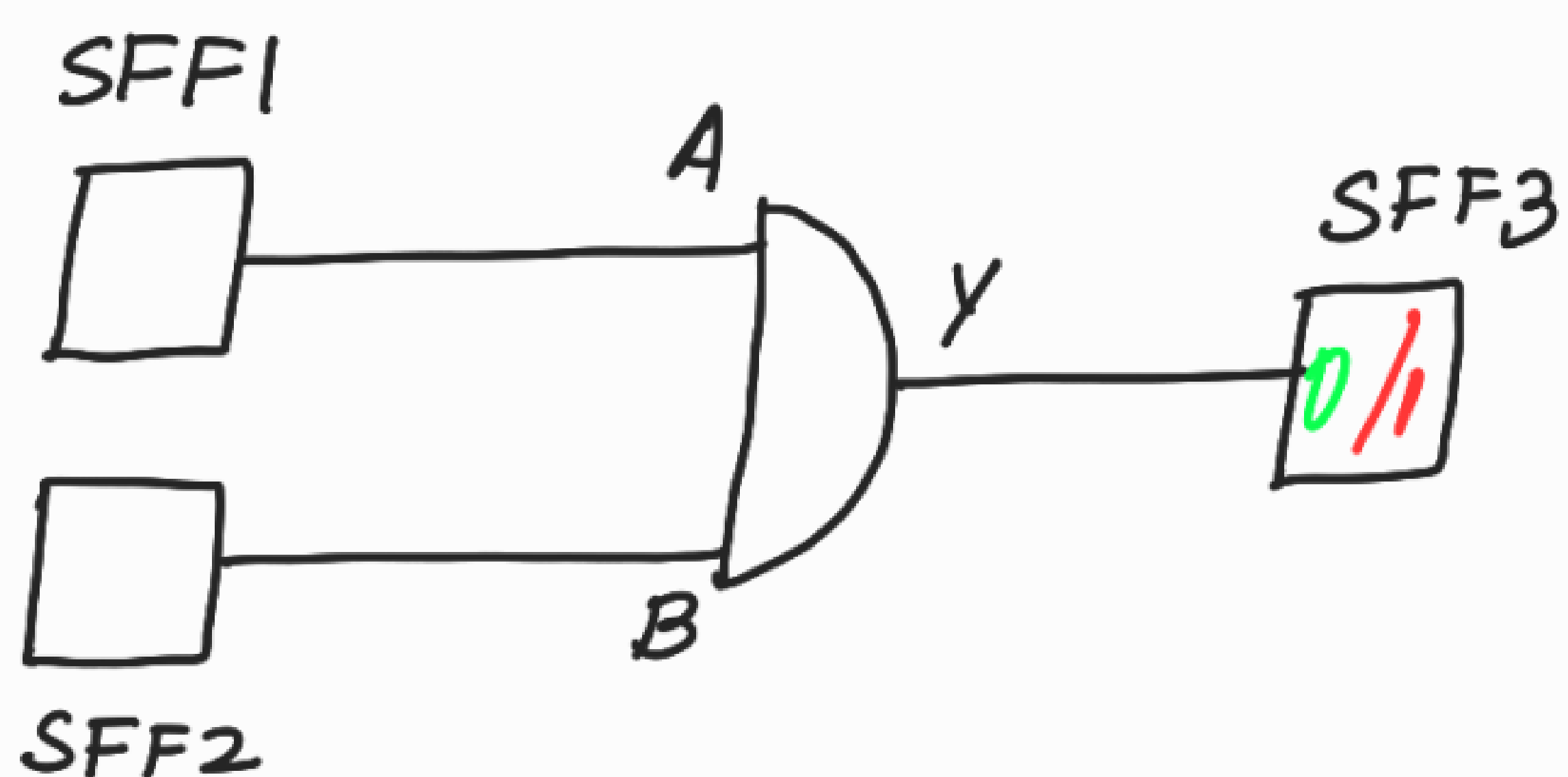
Y slow to fall (Y stf) :-



a pulse



b pulse



c pulse

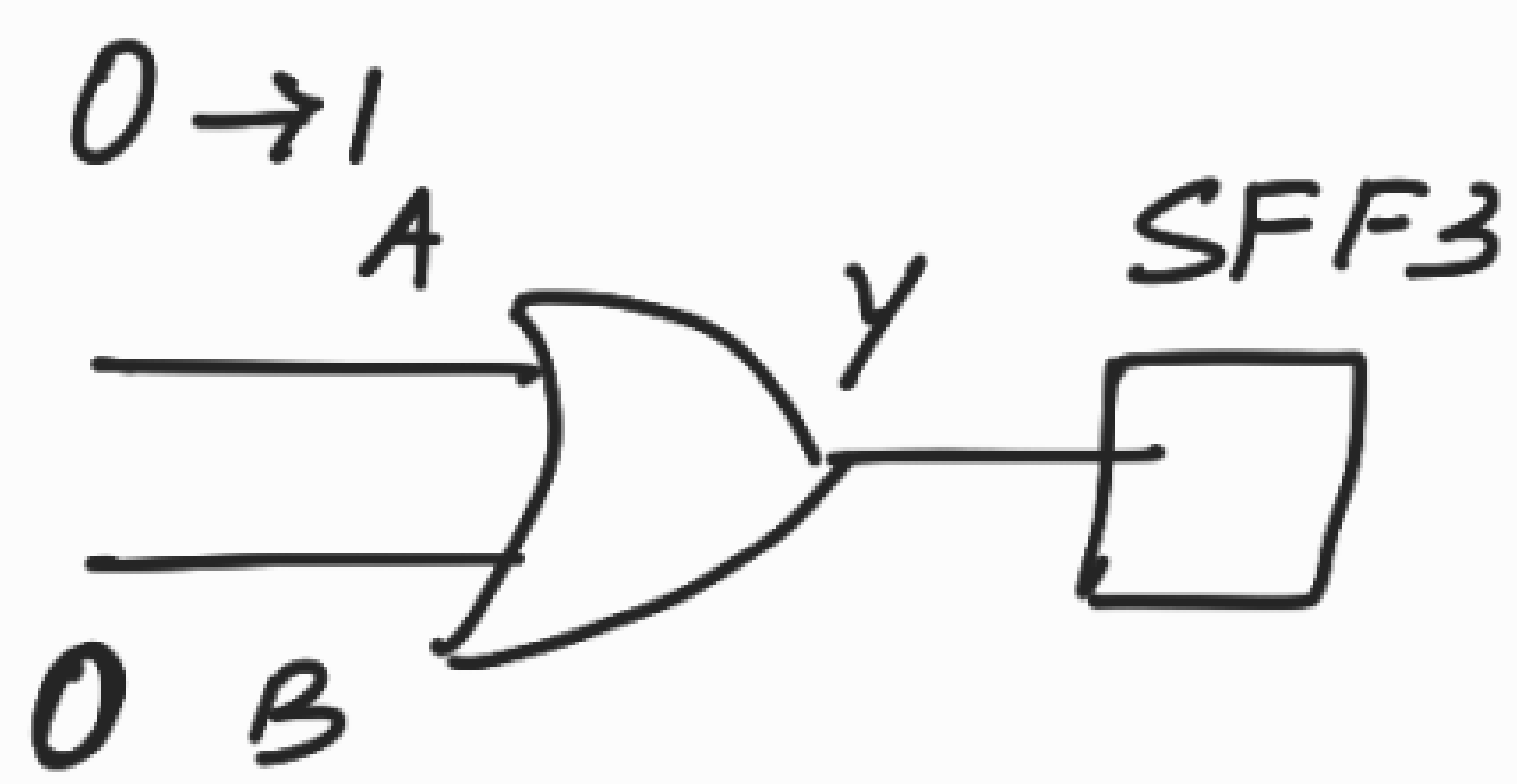
pattern :- $AB \left\{ \begin{matrix} AB \\ 11 \\ 00 \end{matrix} \right\} (OR) \left\{ \begin{matrix} AB \\ 11 \\ 01 \end{matrix} \right\} (OR) \left\{ \begin{matrix} AB \\ 11 \\ 10 \end{matrix} \right\}$

FAULT	PATTERN
A slow to rise	$AB \left\{ \begin{matrix} A \\ 0 \\ 1 \end{matrix} \begin{matrix} B \\ 1 \\ 1 \end{matrix} \right\}$
A slow to fall	$AB \left\{ \begin{matrix} A \\ 1 \\ 0 \end{matrix} \begin{matrix} B \\ 1 \\ 1 \end{matrix} \right\}$
B slow to rise	$AB \left\{ \begin{matrix} A \\ 1 \\ 1 \end{matrix} \begin{matrix} B \\ 0 \\ 1 \end{matrix} \right\}$
B slow to fall	$AB \left\{ \begin{matrix} A \\ 1 \\ 1 \end{matrix} \begin{matrix} B \\ 1 \\ 0 \end{matrix} \right\}$
Y slow to rise	$AB \left\{ \begin{matrix} A \\ 0 \\ 1 \end{matrix} \begin{matrix} B \\ 0 \\ 1 \end{matrix} \right\} (OR) \left\{ \begin{matrix} A \\ 0 \\ 1 \end{matrix} \begin{matrix} B \\ 1 \\ 1 \end{matrix} \right\} (OR) \left\{ \begin{matrix} A \\ 1 \\ 1 \end{matrix} \begin{matrix} B \\ 0 \\ 1 \end{matrix} \right\}$
Y slow to fall	$AB \left\{ \begin{matrix} A \\ 1 \\ 0 \end{matrix} \begin{matrix} B \\ 1 \\ 0 \end{matrix} \right\} (OR) \left\{ \begin{matrix} A \\ 1 \\ 0 \end{matrix} \begin{matrix} B \\ 1 \\ 1 \end{matrix} \right\} (OR) \left\{ \begin{matrix} A \\ 1 \\ 1 \end{matrix} \begin{matrix} B \\ 1 \\ 0 \end{matrix} \right\}$

Final pattern list :-

$AB \left[\left\{ \begin{matrix} 0 \\ 1 \end{matrix} \begin{matrix} 1 \\ 1 \end{matrix} \right\}, \left\{ \begin{matrix} 1 \\ 0 \end{matrix} \begin{matrix} 1 \\ 1 \end{matrix} \right\}, \left\{ \begin{matrix} 1 \\ 1 \end{matrix} \begin{matrix} 0 \\ 1 \end{matrix} \right\}, \left\{ \begin{matrix} 1 \\ 1 \end{matrix} \begin{matrix} 1 \\ 0 \end{matrix} \right\} \right]$

Q:-



B	A	Y
0	0	0
0	1	1
1	0	1
1	1	1

which pattern will detect A stⁿ (slow to rise) fault?

A	B
0	0
1	0

R. Vidisha

Eg 2 :-

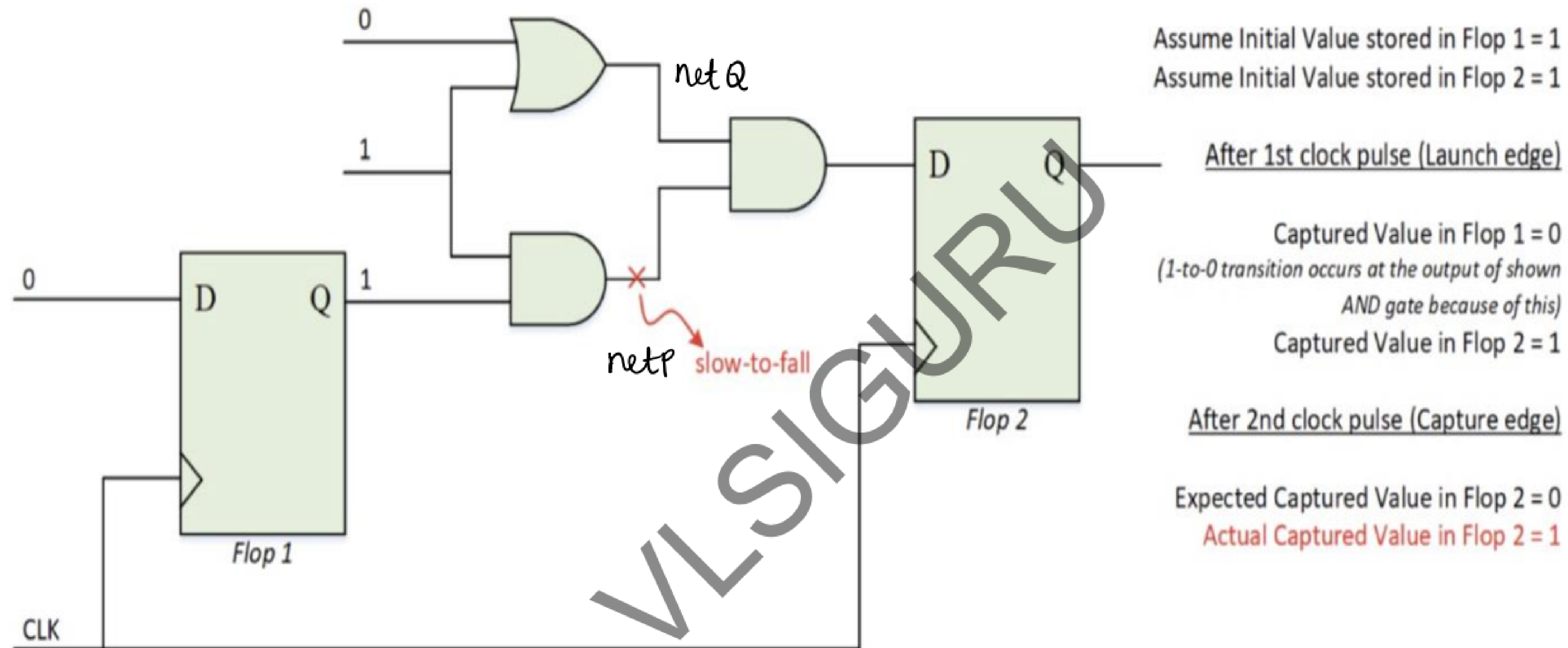
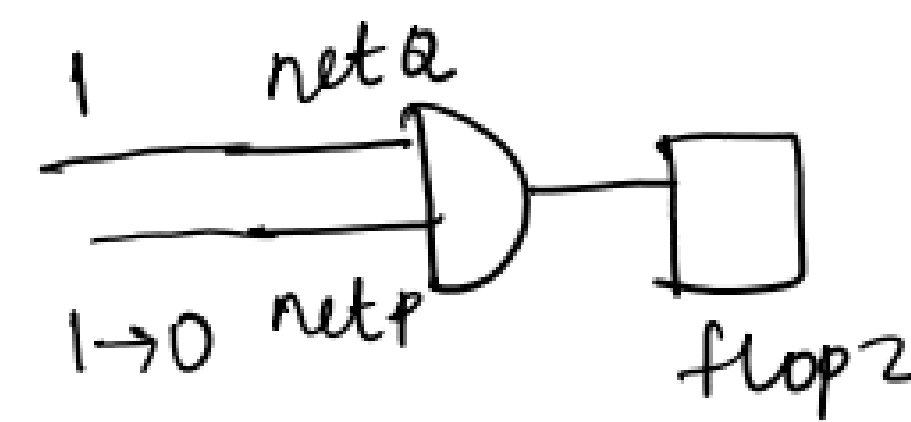


Figure 2: slow-to-fall fault in a circuit

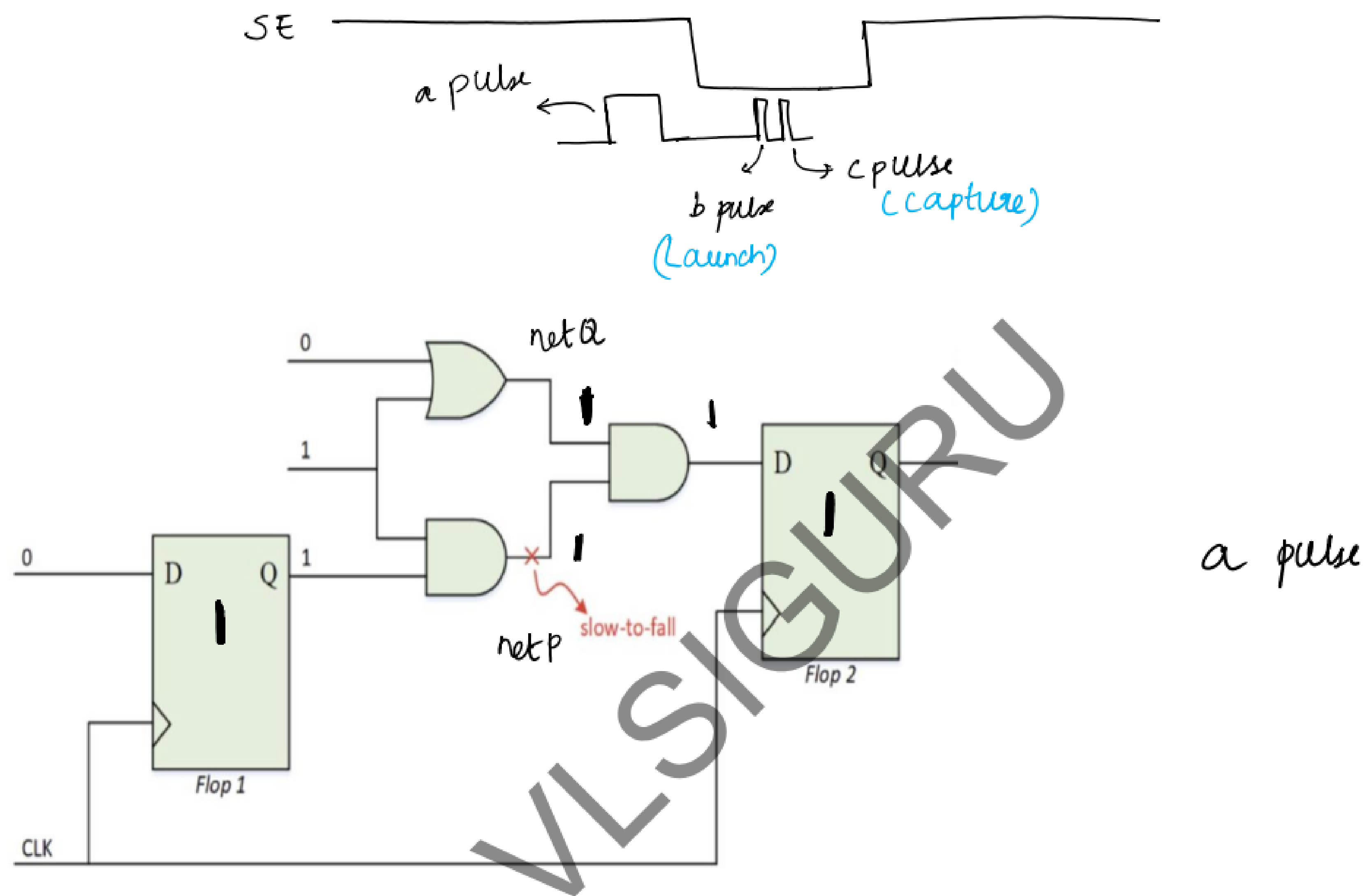


Figure 2: slow-to-fall fault in a circuit

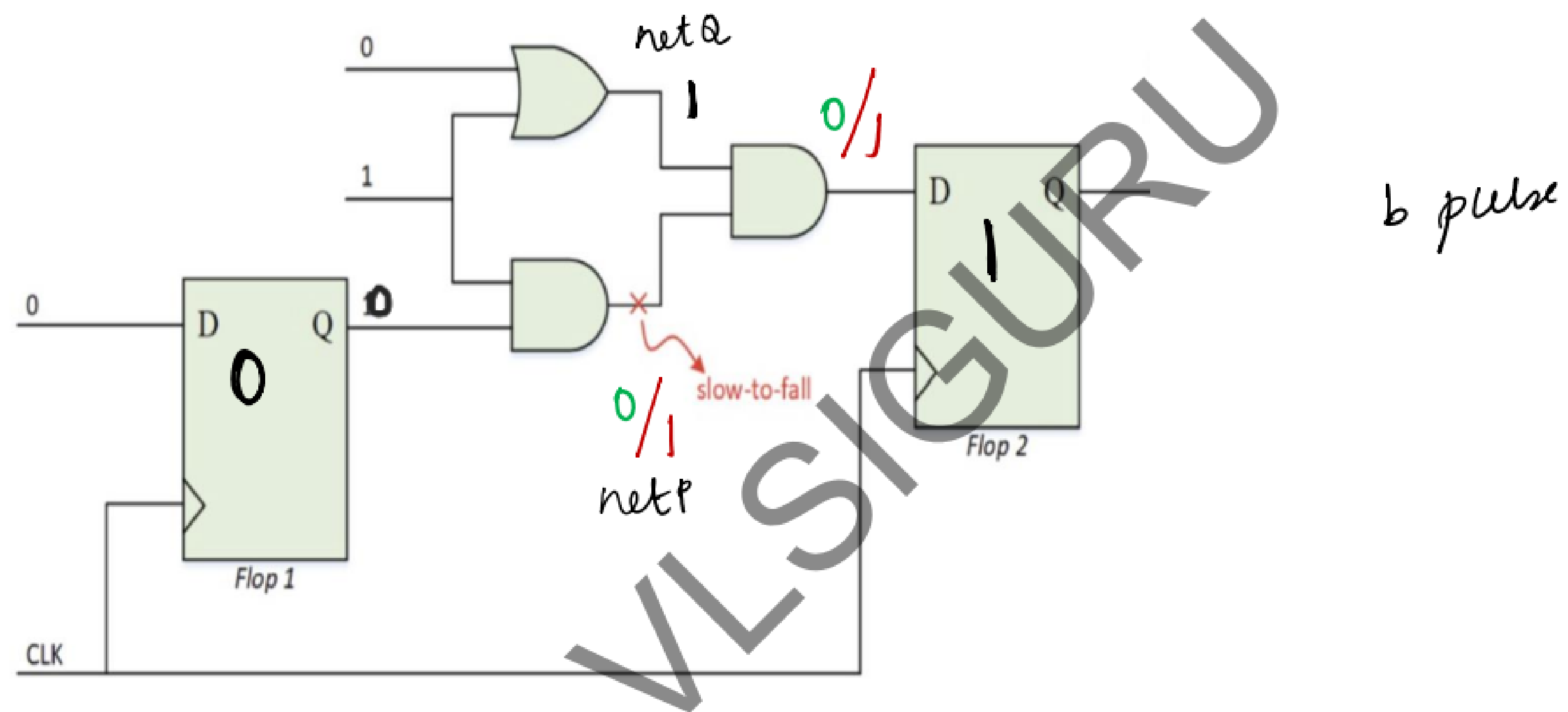
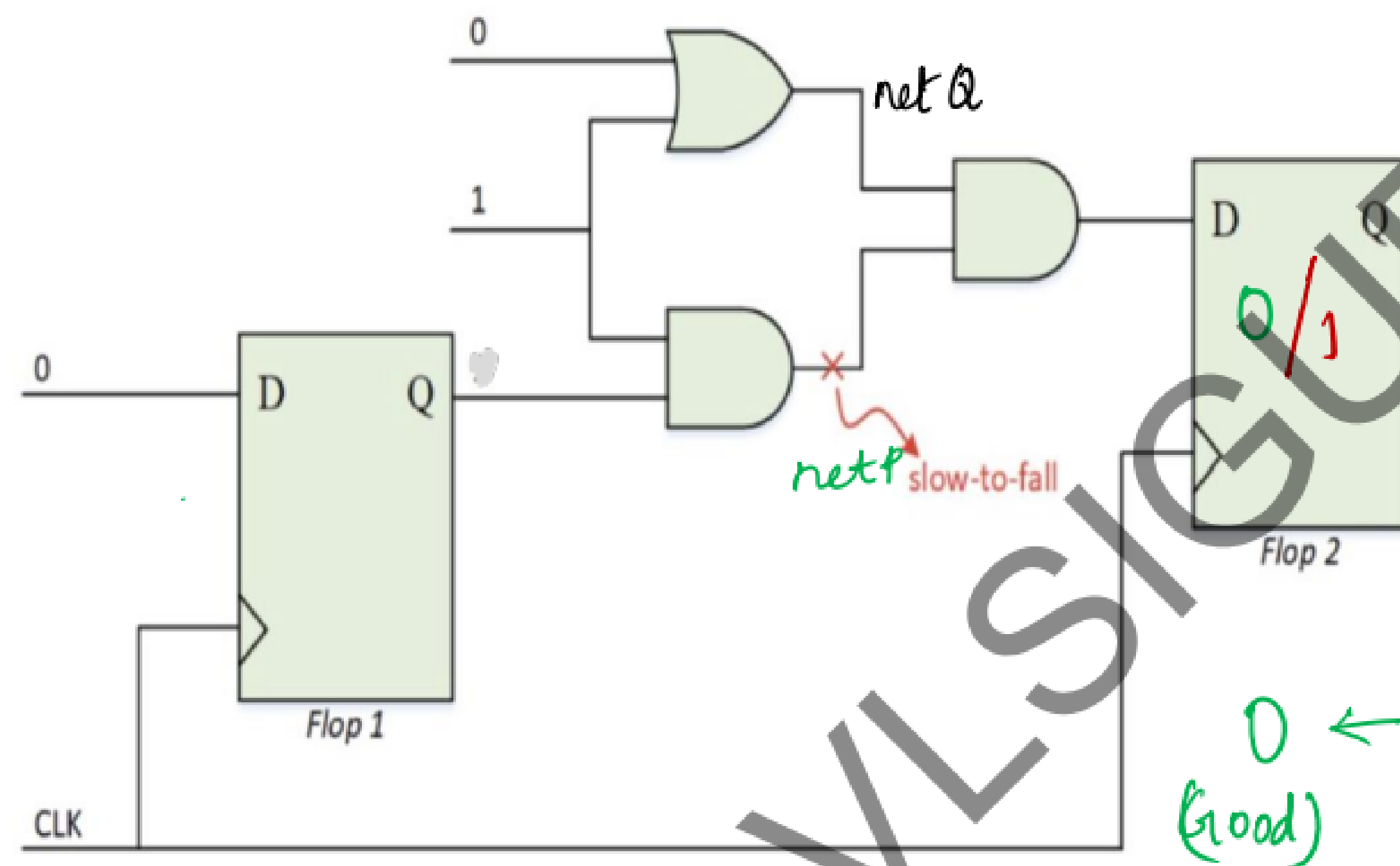


Figure 2: slow-to-fall fault in a circuit



c pulse

0 ← by the time c pulse has come,
(Good) net P has fallen to 0.

1 ← by the time c pulse has come,
(Faulty) net P has not yet fallen to 0.

Figure 2: slow-to-fall fault in a circuit

Waveform:

